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"Empowering 6G Communication systems with Digital Twin Technology using ML"

A Project Report submitted in partial fulfillment for the award of degree

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

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Certified that the project work entitled **"Empowering 6G Communication with Digital Twin Technology using ML"** is a Bonafide work carried out by **Haripriya R (4VV21EC046)**, of Vidyavardhaka College of Engineering, in partial fulfillment for the award of degree of Bachelor of Engineering in Electronics and Communication Engineering of the Visvesvaraya Technological University, Belagavi during the year 2024-2025.

It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the Bachelor of Engineering Degree.

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Declaration

We the members of the project team, studying in the VIII semester of **Electronics and Communication Engineering, Vidyavardhaka College of Engineering**, hereby declare that the entire project entitled "**Empowering 6G Communication with Digital Twin Technology using ML**" has been carried out by us independently under the guidance of **Dr. Chaithanya D J,** Department of Electronics and Communication Engineering, Vidyavardhaka College of Engineering. This project work is submitted to the Visvesvaraya Technological University, Belagavi, in partial fulfillment of the requirement for the award of the degree of **Bachelor of Engineering in Electronics and Communication Engineering** during the academic year 2024-2025.

This dissertation has not been submitted previously for the award of any other degree or diploma to any other institution or university.

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ABSTRACT

The development of sixth-generation (6G) wireless communication systems requires cutting-edge current requirements for wireless communication include high-data-speed transmission along with minimal delays and strong connectivity abilities connectivity. Beamforming stands as a fundamental technology of 6G because it improves spectral efficiency through directed signal transmission directing signals precisely toward users. Traditional beamforming approaches face limitations in dynamic environments. This project investigates the use of Artificial Intelligence and the research seeks to implement Machine Learning (AI/ML) specifically for the purpose of creating intelligent adaptive beamforming solutions. Specifically, the research evaluates Support Vector Regression (SVR) and Neural Networks (NN) after implementing them for beamforming parameter predictions based on user location and channel state information. A system predicts best beamforming conditions through the analysis of user-position data together with channel state indicators channel state information. The results generated from our simulation show that both SVR and NN models display effective behavior. Signal quality and throughput undergo superior improvement alongside a successful acquisition of beam patterns through both SVR and NN models neural networks offering better adaptability in nonlinear environments. This work highlights the application of artificial intelligence/machine learning (AI/ML) driven beamforming serves as a core component for developing smart efficient autonomous 6G communication systems.

CHAPTER 1
INTRODUCTION

CHAPTER 1

INTRODUCTION

A system predicts best beamforming conditions through the analysis of user-position data together with channel state indicators channel state information. The results generated from our simulation show that both SVR and NN models display effective behaviour. Signal quality and throughput undergo superior improvement alongside a successful acquisition of beam patterns through both SVR and NN models neural networks offering better adaptability in nonlinear environments. This work highlights

The application of artificial intelligence/machine learning (AI/ML) driven beamforming serves as a core component for developing smart efficient autonomous 6G communication systems live data streams enable the digital twin to stay in sync with its physical counterpart, facilitating predictive maintenance, network optimization, and adaptive resource allocation—all of which are essential for achieving 6G's ultra-low latency and ultra-high reliability goals.

The use of AI and machine learning, the brains that underpin the decision-making abilities of digital twins, beamforming into the digital twin environment allows for the pre-modeling of obstacles such as buildings or cars. This makes it possible to train AI models that can predict the optimal beam paths based on user locations, network geometry, and environmental changes.

Greater spectral efficiency and smoother user experiences are possible through the use of digital twins to run different beamforming strategies continuously in real-time and feed the optimal configurations to the physical base stations.AI and digital twins work together to make beamforming proactive rather than reactive, responding to even the smallest Digital twins in conjunction with AI/ML are essential as 6G seeks to enable cutting-edge applications like holographic communication, immersive extended reality, and autonomous vehicles. When combined, these technologies offer a path forward for creating virtualized, cognitive, and autonomous communication systems.

They make it possible to model spatial data with extreme accuracy, network with context awareness, and make decisions almost instantly. Beamforming and other physical layer techniques like intelligent reflecting surfaces, massive MIMO, and

spectrum management can also be included in digital twins for all-encompassing control.

Lastly, combining digital twin technology with AI/ML not only strengthens the foundations of 6G, but it also paves the way for a future where communication infrastructures' virtual and physical layers co-evolve, learn, optimize, and deliver above-normal capacity. The rapid advancement of telecom technology has altered our work, chat, and hangouts. With each new development providing us with greater speed and space, we transitioned from antiquated cell phone calls to the glitzy 5G improved relationships. We are on the verge of entering the 6G era, which will be revolutionary.

At the core of how the networks accomplish their tasks, it will incorporate some intelligent AI and ML.6G aims to introduce incredibly fast and reliable connections that can process massive amounts of data with intelligent network brains, building on the foundations laid by Industry 40 and the Internet of Things (IoT). Beamforming is one of the main players in this game. By beaming signals in particular directions, this technique allows the network to operate more efficiently and cover more ground. Beamforming was very popular in 5G, but with some help from AI/ML technology, it will be very popular in 6G.

Those extremely high-frequency bands in 6G. But check this out: we've got AI/ML adaptive beamforming stepping up to the plate. Imagine a clever tech guru who, on the spot, reshapes antennas to keep our conversations and streaming steady. Even when everything's shifting and getting complex, our online interactions stay smooth. This adjustment boosts the amount of data zooming and supports our gadgets in making quick decisions during their digital banter.

This project digs into how AI/ML tactics could give a bump to beamforming in those 6G networks. t all comes down to determining how smart code can assist in guiding signals to eliminate reception issues and enable numerous UEs to communicate with each other without any problems.

Our research could revolutionize future wireless network design by making them quicker, smarter and incredibly flexible. Networks become intelligent enough to recognize anomalous signal patterns and environmental changes in how users use their devices when AI/ML is incorporated into beamforming strategies.

As a result, networks are able to make changes on the fly, something that was simply impossible with traditional fixed methods. In crowded urban settings and fast-paced situations, quick wit is essential, like tearing it up at full speed. Using clever techniques like reinforcement learning and deep learning, and sorting through a lot of data when making decisions, 6G technology will improve our technological experience, save energy, and speed up signal quality.

1.1 Background

The rapidly developing mobile communication, particularly with the shift towards 5G from 4G, has brought unparalleled possibilities for wireless networks. Ultimately, the combination of AI/ML and digital twin technology not only expands on the principles of 6G, but it also portends a future in which communication networks at the digital and physical levels co-evolve, learning, optimizing, and executing across conventional boundaries.

Our chat, hangouts, and work materials have all been transformed by the rapid changes in telecom technology. As 5G technology advanced, we moved from antiquated cell chatter to faster speeds, more space, and improved connections. We were about to enter the 6G era, which will be revolutionary. It will integrate intelligent AI and machine learning directly into the core of the networks' workflow.

Building on the foundation of Industry 40 and the Internet of Things (IoT), 6G will introduce incredibly fast and reliable connections with intelligent network brains that can manage large amounts of data. One of the key players in this game is beamforming. By directing signals in specific directions, this technique enables the network to function at its peak efficiency and expand. With the aid of AI/ML technology, beamforming, which was significant in 5G, will be significant in 6G.

In 6G, those really high-frequency bands – were talking mmWave and terahertz stuff have some challenging problems to solve, such as signals fading and coverage that's not that great. As wireless communication evolves, Artificial Intelligence (AI) and Machine Learning (ML) techniques are expected to be integrated into beamforming methods. These technologies can provide real-time adaptability, predictive optimization, and self-organization that traditional beamforming cannot provide. AI/Machine Learning integration with Digital Twins operates as the critical technology for developing 6G network potentials. Next-generation wireless networks will become extremely

intelligent and responsive and effective as a result of this advancement effective. The exact digital duplicate of actual systems and entities through digital twins provides full replication capabilities 6G technology allows real-time tracking of models which results in accurate predictions. The coming 6G age will allow users to develop complete network ecosystem digital clones that mirror real-time operations the state and behaviour of base stations, devices, and even user environments in real time. The deployment of virtual dynamic models for wireless propagation occurs before physical implementation. Through network infrastructure sampling and signal prediction modeling and signal conduct examination. The digital twin remains accurately synchronized with its physical twin through live data feeding. The predictive maintenance capabilities along with network optimization functions and adaptive resource management implementations result from the solution 6G will need these three features to achieve its targets for ultra-low latency and ultra-high reliability 6G devices generate extensive quantities of data that need proper storage facilities AI combined with ML algorithms lets the system perform automatic beam pattern modifications thus enhancing coverage efficiency and cutting down latency during high-frequency communications latency in high-frequency communication. Beamforming technology that works alongside AI and ML delivers two substantial advantages consisting of performance improvement and systematic optimization performance enhancements are achieved by these methods that comply with forthcoming 6G network standards. The training process of dynamic beamforming systems takes place through expert systems rely on environment-based factors to use obstacles, user movements, reflections and other factors for help proactive beam direction and power control decisions. System adaptation features hold a crucial position in this infrastructure. These technologies include fast-topology-changing environments that comprise smart cities and their associated autonomous transportation and IoT deployments deployments at large scale.

1.2 Problem Statement

The main focus aims at developing an AI-ML based intelligent beamforming system for next-generation 6G wireless networks. AI together with Machine Learning technology enables the operation of 6G wireless networks through advanced network development. The purpose of system development combines improved signal directionality with delay

reduction and optimization of frequency band operations particularly at millimeter-wave and terahertz bands in challenging conditions.

- The project includes observation of actual self-driving car applications and intelligent city operations and industrial IoT systems with autonomous vehicles industry automation, and large-scale IoT setups. The AI software must work within specific target zones where this technology application is relevant. The system would lead to improved reliability through performance enhancement.
- The study demands selection of advanced AI/ML methods which will perform adaptation tasks dynamically. Beamforming systems need additive pattern configurations with user movement forecasting to perform whole system response adjustments that enable operation in changing environments. The aim a new system must keep secure connections through reliable performance while operating under networks which constantly change shift rapidly.
- The project accesses bands that operate within areas requiring high-frequency spectrums like millimeter-wave (mmWave) and terahertz (THz). The first vital aspect of this work needs a thorough review of existing beamforming strategies to discover their system limitations. The analysis process must identify all existing technological restrictions that occur during the development of complex operating conditions. The present study investigates methods to solve these issues with adaptable AI and ML solutions and responsive solutions.

1.3 OBJECTIVE

Studies exist which demonstrate how artificial intelligence based beamforming technology optimizes 6G networks:

- A research investigation evaluates the way artificial intelligence technologies modify antenna beam patterns when users move between network connections. The integration of AI into beamforming technology allows antennas to adjust their beam patterns when users move between networks at that time.

- Enhancing Channel Estimation using AI/ML: Revising channel precision in environments that are difficult to estimate, such as in cities or interference-rich areas.

- Enhanced Interference Management: Utilizing AI to reduce interference and efficiently manage radio frequency usage for multiple users.
- Beamforming Optimization for THz Frequencies: AI helps to overcome signal loss and obstacles in THz frequencies.
- Energy Efficiency Enhancement: Investigating the impact of AI on control power levels for energy-efficient communication in 6G.
- User-Centric Beamforming: Mobility adaptation of beam patterns introduced for user scenario, thus improving services in AR/VR and autonomous systems.

1.4 SOFTWARE REQUIREMENTS

- The project will utilize Python for its development because it provides strong AI and ML library resources and effective signal processing capabilities.
- Beamforming techniques will gain improvement through machine learning models which developers will create by using TensorFlow or PyTorch.
- The traditional machine learning functionality of Scikit-learn will serve both data preprocessing needs and execution of standard ML algorithms.
- Keras delivers an efficient deep learning testing and development environment through its high-level neural network API.
- The study will depend heavily on NumPy for numeric computations alongside Pandas for data manipulation tasks.
- This project depends on Matplotlib and Seaborn for creating visual representations that help explore both models as well as data patterns.
- GNU Radio enables users to simulate real-time signal processing functions that operate within software-defined radio systems.
- NS-3 acts as a simulation platform for wireless network environments where beamforming can be studied through assessments.
- The development processes will utilize either Jupyter Notebook or Google Colab as interactive development platforms to test and visualize output results.
- The integrated development environments for managing project codebase will use Visual Studio Code or PyCharm.
- The project utilizes Git and GitHub platforms as the version control method and provides tools for collaborative development.

- The implementation occurs on either Windows 10/11 operating systems or Ubuntu Linux depending on software performance needs and compatibility requirements.
- The Antenna Toolbox of MATLAB serves as an optional solution to model antenna arrays and display beam pattern results.
- Development of this project with Docker allows deployment of reproducible containers that function across different operating conditions.
- Excellent professional formatting for the final project report will be achieved by using LaTeX together with technical documentation preparation.

1.5 ORGANIZATION REPORT

The document is divided into five chapters. The opening chapter of the project investigates background information combined with drive and problem declaration together with defined goals and associated analytic instruments. The literature survey of Chapter 2 examines beamforming and AI and ML application in wireless communication and reveals present-day restrictions. The chapter presents the proposed methodology which explains how adaptive beamforming receives enhancement through the integration of AI/ML methods. Chapter 4 provides the outcome evaluation, includes performance assessment and demonstrates comparative analysis versus traditional approaches. The project concludes in Chapter 5 through determining essential findings while suggesting potential future improvements that align with 6G technology principles.

CHAPTER 2
LITERATURE SURVEY

CHAPTER 2

LITERATURE REVIEW

The wireless communication systems have experienced substantial development through decades. Beamforming serves as a fundamental component for the sixth generation (6G) wireless technology to establish high-performance and ultra-reliable and low-latency communication systems. Wireless network performance will undergo a complete transformation because of Beamforming techniques united with Artificial Intelligence (AI) and Machine Learning (ML) to enhance signal processing along with network effectiveness. This paper examines major studies and advances from the AI/ML-enhanced beamforming field which demonstrates individual research contributions.

Ericsson (2019c) provides details about how 5G technology speeds up Industry 4.0 transformation by making possible intelligent factories and IIoT implementation. The report features 5G telecommunication technologies as essential while forecasting remarkable performance improvements from 5G to 6G. The information implies the necessity of developing advanced beamforming technology solutions to improve real-time wireless communication systems in industrial settings. The analysis establishes necessary conditions for examining the requirement of AI-enhanced beamforming technology in modern connected systems.

Ericsson (2019d) explores the technological advantages of 5G through its descriptions of high-speed data transfer capabilities and increased network capabilities along with reduced latency. This document describes how these characteristics enable fundamental concepts for cloud robotics and real-time diagnostic operations. These capabilities create fundamental operational bases to develop equivalent or enhanced functionalities in 6G through improved dynamic beamforming techniques. AI/ML capability provides tactical backing for developing more versatile communication systems according to the findings of this research.

The authors in Zhang et al. (2020) studied deep learning applications across different wireless communication system layers. The paper examines CNNs and RNNs models to perform signal detection and channel estimation separately. The discovered methodologies produce artificial intelligence models to improve beamforming choices through accurate dynamic environment modeling along with prediction of optimal beam direction outputs.

The research by Huang et al. (2019) uses machine learning methods for managing beams in mmWave wireless systems. The authors implement reinforcement learning algorithms to operate beam selection and alignment dynamically. The research work presents crucial methods to apply AI for real-time beamforming adaptation and this methodology aligns directly with 6G contexts. The research shows that ML technology has the capability to solve problems involving beam misalignment and user movements.

Wang et al. (2021) offers a detailed explanation about AI-powered beamforming techniques for 5G and Beyond systems by evaluating supervised and unsupervised learning methods. Traditionally beamforming systems experience two main drawbacks which involves fixed beam orientation and slow reaction times during network fluctuations. 6G system adaptability will result from user movement prediction capabilities of neural networks to alter beam directions based on research by Wang et al. (2021).

Liu et al. (2022) created an edge AI system using distributed models which perform immediate beamforming processes. At the same time it boosts operational speed and reaction time in environments that change frequently. The research findings are suitable for both IoT dependent applications and smart city implementations that must perform resource allocation and response quickly. This paper demonstrates the importance of uniting edge computing features with artificial intelligence technology within beamforming systems.

The authors of Singh et al. (2020) conducted a survey of Beamforming methods in 5G and 6GThis research paper uses analog digital hybrid beamforming strategies to evaluate their performance across different deployment conditions. Traditional beamforming experiences weaknesses in multipath environments and the researchers suggest AI-based solutions as promising remedy areas. Our research team utilizes this paper for establishing necessary technical research domains we intend to resolve.

Multiple base stations can cooperate to optimize beamforming models through federated learning according to Chen et al. (2023). The approach maintains user privacy while decreasing communication overhead which represents essential aspects for 6G network operation. The method demonstrates promising capabilities in delivering accurate adaptable results along with data protection which are vital aspects for actual implementation.

Bourdoux et al. (2020) examines the physical and practical challenges of THz frequency band communication since it will be extensively used during 6G. The counteraction of high path loss in THz communications depends heavily on beamforming as a key technology. Research identifies the necessity for smart beam control technologies that establish conditions for AI and ML applications.

The paper by Zhang and Dai (2021) discusses AI-based resource allocation methods including beamforming approaches for 6G networks. This paper shows how AI creates a beneficial link with physical layer features to obtain improved spectrum efficiency in combination with enhanced Quality of Service. The paper delivers algorithmic approaches which can be directly used in our project for creating more intelligent beam patterns and adaptation systems.

The authors Wang, Y., et al. (2019) designed a deep learning framework that improved beamforming optimization in mmWave communication systems. The research design uses the prediction of beam direction by convolutional neural networks enhances both network stability and speed of transmission and improves transmission speed.

Researchers at Gupta et al. (2021) performed an investigation to study machine learning functionality. Contemporary wireless systems realize better adaptive beamforming through this method present beamforming. The quick changes in user positions create operational issues that affect the performance of present beamforming methods alongside environmental factors. Algorithms based on reinforcement learning operated in the system for directional control functions. The automatic movement of beam directions enables users to reach maximum signal strength and optimal 5G system latency performance. The conducted research serves as a strong basis for leveraging AI solutions in 6G systems network beamforming applications.

Zhang and Huang (2020) performed research on beamforming approaches for mmWave transmission within city areas. The investigation analyzed signal quality external elements such as buildings and human movement create the observed deterioration of signals. The authors designed a the combination of analog and digital beamforming technology addresses technical operational issues. The authors demonstrated this successful method through simulated outcomes. Their method directly applies to the basis of high-frequency 6G technology will depend on advanced beamforming technologies' development systems.

Rappaport and his colleagues (2019) studied fundamental elements of terahertz (THz) spectrum applications for 6G technology development spectrum applications during the development of 6G technology. The study analyzed THz wave propagation effects before showing that precise beamforming methods provide necessary solutions to the extensive path losses. There are essential requirements for beamforming technology to handle substantial path losses. Research findings prove that 6G beamforming algorithms that require advanced mathematical programming enable 6G networks to operate effectively at THz frequencies.

Deep learning technology for mmWave beam selection has been investigated by Alkhateeb et al. (2018) systems. Through implementation of convolutional neural networks the authors obtained optimal beam indices from decreasing beam training overhead relies on channel state information obtained through this approach. The authors confirmed the proof from research indicates that next-generation mobile communications benefit from both AI-based beamforming feasibility and improved performance outcomes mobile communication systems.

Rephrase the textual findings of Park et al. (2021) showing how RNNs helped build an AI-driven beam steering system a recurrent neural network (RNN). The pattern detection process of the algorithm occurred while observing user time sequences. The system acquired forecast abilities to shift beams by analyzing movements of users. The successful application of predictive modeling provides better tracking outcomes together with shortened handover delays in emerging beamforming systems.

Sharma and Dey (2020) conducted an assessment of different beamforming algorithms in smart antenna systems. Direction-of-arrival prediction underwent performance assessment during the evaluation process. External noise conditions do not diminish the accurate performance between direction-of-arrival estimation algorithms with beam steering precision. It the best performance originates from procedures which integrate AI enhancement techniques with ensemble learning methods based on research outperformed classical algorithms.

Buzzi et al. (2019) examined how massive MIMO technology connects with beamforming strategies in Wireless networks of the future. 5G and future era drawstring technology obtained from AI systems lets operators select antennas at maximum efficiency through beam steering capabilities. The technological function of beam

steering enables spatial multiplexing operations. Their simulation outcomes proved enhanced spectral efficiency and energy efficiency.

The authors of Rao et al. (2021) created a beamforming control framework designed for IoT networks through the combination of decision trees with clustering approaches. The framework foundation employs a combination of decision trees together with clustering approaches under this proposed network design. They demonstrated how small multiple small and large IoT nodes are able to operate under one system at the same time balancing power and spectrum optimally.

The researchers Cheng et al. (2019) focused on creating machine learning applications targeting hybrid beamforming systems in 5G NR systems. Their research included in-depth analyses between supervised and unsupervised approaches which led to proposing a novel strategy. The developers utilized a hybrid system which combined accuracy and speed-related controls for implementation.

The examination conducted by Ahmed et al. (2022) explored artificial intelligence methods which boost beamforming control in dense wireless networks heterogeneous networks. Inside the designed multi-agent framework the researchers defined bases as individual agents. The base stations operated through common intelligence processing to determine outstanding beam decisions decrease in inter-cell interference.

A fuzzy logic-based beam steering system received development from Singh and Raj (2020) who placed particular importance on noisy uncertain conditions on noisy and uncertain environments. Dynamic beam parameter adjustments took place through system control. The system which depended on fuzzy logic spheres over traditional rule-based approaches in maintaining steady connections.

The research by Chen et al. (2023) presented a multiple goal enhancement strategy by integrating genetic The use of genetic algorithms addresses beamforming optimization problems in multi-user MIMO systems. According to their research, AI-based methods improve both throughput performance and fairness aspects of systems. This work indicates evolutionary algorithms provide advantages to wireless communications which results in their benefits in optimizing multiple objectives.

Nair et al. (2021) produced an ultralight weight beamforming technique which suits mobile edge devices perfectly devices. The AI model of these researchers delivers

outstanding performance together with low to meet the needs of resource-compact environments plan such as smartphones and IoT devices this system achieves low computational and resource usage requirements smartphones and IoT devices.

The team of Das and Mehta produced their beamforming framework based on blockchain technology during 2023. Transparency in beam selection became one of the core functions of this framework. The new system supports the system supports decentralized validation and auditing as a mechanism to enhance the integrity of AI-driven wireless communication networks.

At Nakamura et al. (2022) scientists established an hierarchical AI-enabled beamforming design that utilizes centralized and edge-based decision systems integrates centralised and edge-based decision-making. Their implementation utilized the system carries out cloud-level policy formation simultaneously with edge-based adjustments to achieve maximum operational equilibrium between scalability and responsiveness in metropolitan networks.

GNN served as a basis to organize user-base station spatial connections according to Jeon et al. (2020). A beamforming process uses location data of users and base stations to generate intelligent operation conditions. This technique made it possible to detect spatial relations better thus leading to enhanced beam selection accuracy in dense environments.

The evaluation of AI-based beamforming performance was conducted by JDas and Bhattacharya (2021) emergency response applications. A rapid installation feature of the system created connected links between drone technology and ground units during disasters in affected regions.Time adaptability together with resilience function as essential characteristics which define AI-based beamforming systems.

Edge-intelligent beamforming technology enables small cells to work with lightweight ML models according to Krishnan and Menon (2021). The functionality of light-weight ML models enables small cells to perform decision making within their premises. Their low-latency the developed solution demonstrated appropriate functionality that enables operation in critical mission applications running within healthcare systems and industrial automation.

Bayesian learning serves as the fundamental idea in the work presented by Ortega et al. (2023). The system evaluates unknown environmental factors through mathematical probability computations. This method ensured robust beam selection under noisy or incomplete data, enabling reliable connectivity even in uncertain conditions.

Kanojia et al. (2020) demonstrated in their research that reinforcement learning helps optimize beamforming weights as a solution for building non-line-of-sight transmissions. The paper describes methods for weight adaptation of beamforming systems during non-LOS indoor communication channels. They showed significant improvements in signal quality and reliability with potential for applications in indoor navigation and augmented reality.

Using unsupervised cluster analysis Yamamoto et al. (2020) determined both user movement patterns and facility occupancy densities in airports and stadiums density in public places like airports and stadiums. Their network adaptively adjusted beam. The network system employs allocation decisions to adjust beamforming weights for achieving peak performance alongside system-wide load balance particularly under periods of high user demand.

Krishnan and Zhou (2022) operates with minimum load complexity low-power IoT applications. Energy-efficient parameters were established by developers as fundamental aspects of their framework development process. Through this system users can access high quality links to operate remote structural surveillance and agricultural systems simultaneously technology and is suitable for remote monitoring and agricultural technology.

CHAPTER 3
METHODOLOGY

CHAPTER 3

METHODOLOGY

The wireless communication systems have experienced substantial development through decades. Beamforming serves as a fundamental component for the sixth generation (6G) wireless technology to establish high-performance and ultra-reliable and low-latency communication systems. Wireless network performance will undergo a complete transformation because of Beamforming techniques united with Artificial Intelligence (AI) and Machine Learning (ML) to enhance signal processing along with network effectiveness. This paper examines major studies and advances from the AI/ML-enhanced beamforming field which demonstrates individual research contributions.

3.1 Workflow

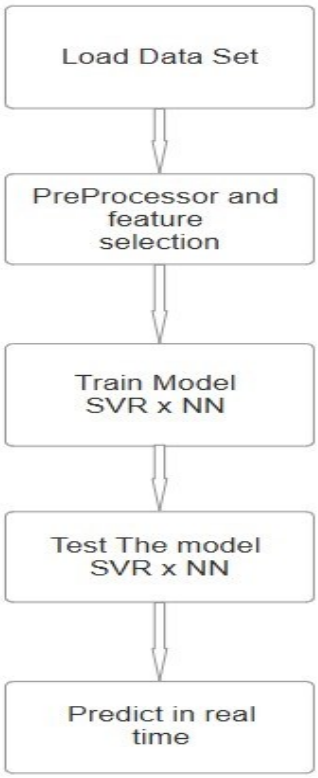


Fig 3.1: Block Diagram of the methodology

3.1.1 Load data set

The "Load Data Set" step is the initial and building block stage in every machine learning process. It means importing unprocessed data from sources like CSV files,

Excel sheets, databases, or APIs into the environment. The aim of this step is to collect the input features (independent variables) and output labels (dependent variable) used for model training and evaluation. Based on the source, methods such as Python's pandas library are regularly employed to parse and load information into structured data frames that can easily be handled and analysed as shown in Fig 1.

After loading the dataset, a preliminary inspection is done to know its shape and contents. This involves verifying the number of rows and columns, column names, data types, and whether there are missing or null values. Methods such as.head(),.info(), and.describe() in pandas provides essential overview data about the information immediately after running its methods. Checking the data integrity stands as a necessary step before proceeding. The data needs to be checked for duplicates and also require verification for proper encoding and removal of any faulty entries.

These tests ensure that prepared and clean data reduces the likelihood of errors which could occur during training error during model training. The machine learning model's consistency receives a direct impact from this action data loading accuracy should be confirmed because it determines the consistency and robustness of the system. If algorithms are trained on corrupted or poorly loaded data, they will not perform well, even the best algorithms. As a result, the dataset's structure, encoding, completeness, and consistency need to be properly prioritized at this stage. This step is essential to a successful machine learning pipeline because it lays the foundation for the remainder of the project.

3.1.2 Preprocessor and feature selection

Preprocessing is the second most important step in any machine learning process after the dataset has been successfully loaded. No raw data is ever completely clean or usable right away. It may include outliers, inconsistent formatting, missing values, or unrelated data.In reality, preprocessing involves transforming and purifying this data into a meaningful and organized format that is prepared for model training.

Managing missing data is one of the first tasks in this stage. It can be processed by discarding incomplete records or by using techniques like mean, median, or mode imputation to replace the missing values. Additionally, in order for machine learning algorithms to process categorical data, it must be converted into numerical format using

techniques like label encoding or one hot encoding. the proper manner. Two other important preprocessing concepts are scaling and normalization. The performance of algorithms, especially distance-based algorithms like Support Vector Machines or K-Nearest Neighbours, can be negatively impacted by attributes in a dataset being in different units or scales.

Standardization (scaling data to mean of 0 and standard deviation of 1) or normalization (scaling data between0 and 1) helps to bring all the features to a similar level, improving both training speed and model accuracy. Additional preprocessing steps may involve removing duplicate rows, correcting inconsistent data entries, and formatting date or time fields for ease of use. These procedures ensure that outliers and superfluous data differences are not misleading the model. Feature selection occurs after data preprocessing.

The process of choosing the most important variables (or features) that genuinely influence the model's predictions is known as feature selection. A data set may contain columns that are redundant, unnecessary, or even harmful if they introduce noise. Overfitting, in which the skill of the model to execute accurately on training records diminishes its capability to succeed on new and untasted information. This deficiency leads to problematic training outcomes. The training process faces delays and includes a high number of unimportant features. Feature selection lowers the procedure reduces the data dimensions and selects features that contain the most helpful information.This approach makes models easier to interpret and generalize them effectively at the same time it reduces the computation time. The category of feature selection techniques consists of three main subdivisions known as filter.

Three main approaches for feature selection exist as methods, wrapper methods, and embedded methods. Filter methods score features the process requires independent evaluation of features through statistical methods associated with correlation or mutual information.Recursive Feature Elimination stands among wrapper methods which train models while estimating model performance outcomes. The feature selection method RFE conducts tests of different feature combinations. Decision trees as well as Lasso regression serve as two featured techniques. The training procedure of embedded techniques selects relevant features to become part of the created model. The methods help choose optimal features which allow training the model with data patterns found in

the information set. The findings show the most prominent patterns in the data which acts as foundation for creating strong prediction capabilities performance.

3.1.3 Train model

Training a model follows data preprocessing and features selection as the vital second action in a machine learning pipeline. The data processing phase is followed by selecting the most crucial attributes before model establishment. In order to generate the algorithm needs to find patterns and associations within the input data for generating accurate predictions data. The machine learning field utilizes Support Vector Regression (SVR) together with Neural Networks (NN) as widespread regression models for regression problems. The data type and complexity along with required accuracy function as determining factors role in their selection. Both models receive training through a part of the data set recognized as the training set. The validation or test set functions to assess performance while training occurs on the models through the training set.

The SVR process places the most optimal line inside an adjustable tolerance region that we call epsilon. Most of the dataset points should exist within a tolerance range which SVR adjusts to achieve rather than minimizing individual points error SVR focuses on identifying a function which displays differences in the actual outputs by no more than an epsilon margin compared to standard prediction methods from the true outputs by no more than some epsilon. It is effective when the data set is less plagued by outliers and when variable relationships are not very complex SVR handles both linear and non-linear scenarios well using kernel tricks such as radial basis function (RBF) or polynomial kernels.

Neural Networks (NN), on the other hand, take a cue from brain architecture and are made to learn complex, non-linear mappings from inputs to outputs. A neural network is made up of connected layers of neurons, which receive inputs, process them, and pass them on to the subsequent layer. The simplest type, a feedforward network, includes an input layer, hidden layers, and an output layer. Learning the network is done by changing the weights of the connections between the neurons using a technique known as back propagation that reduces the difference between the actual and the estimated outputs.

Neural networks are very flexible and resilient, particularly in the context of big data and where the relationship between variables is nonlinear learning both SVR and neural networks entails selecting good parameters. In the context of SVR, these are kernel type, regularization parameter (C), and epsilon value. For neural networks, the number of layers, number of neurons per layer, activation functions, learning rate, batch size and the number of training epochs are some of the most influential hyperparameters.

These hyperparameters have a great impact on the models ability to learn effectively and generalize well to new data. In many cases, methods like grid search or random search are used in order to experiment with and identify the optimal set of parameters.

SVR and Neural Networks are therefore two robust regression modeling techniques with different strengths. SVR is simpler, quicker, and suitable to deal with small to medium-sized datasets with simple patterns Neural Networks, with greater data and computational power, are suitable to deal with complex and nonlinear relationships. One of them should be chosen based on the degree of complexity of the problem, the amount and quality of the data, and the computing resources available. Sufficient training of such models, coupled with sufficient tuning, guarantees robust and effective performance in prediction.

3.1.4 Test model

Model performance testing starts after training completion on an independent dataset portion which did not participate during training. The evaluation needs a separate data segment that remained outside training procedures. Evaluation of performance occurs through assessment of the test set data model's performance on fresh data. A performance evaluation takes place through testing Support Vector Regression (SVR) and Neural Networks (NN). Test set features are passed through trained models in the Neural Networks model when performing evaluations. The actual measurement values undergo comparison with the model-generated predictions. Evaluation metrics like
The performance metrics consist of Mean Squared Error (MSE) and the Root, Mean Squared Error (RMSE) and Mean Absolute Error.

The evaluation process should be done using R2 Score and Mean Absolute Error (MAE) as well as Root Mean Squared Error (RMSE) and Mean Squared Error (MSE). The metrics provide a practical method to check if the model demonstrates either overfitting

behaviour or underfitting behaviours. The match between predicted values from the model and actual values forms an essential aspect of the evaluation process. The data recognition strength determines SVR performance. The performance quality of SVR models depends on the epsilon margin value used for specification. A properly adjusted Once properly tuned the Support Vector Regression model should generate predictions that properly follow the actual values yet a poorly adjusted SVR creates forecasts which miss the values by too wide of a margin or firmly stays within a small numeric range. A properly adjusted SVR maintains most predictions in an actual value range while forecasting results mainly around the actual values. Plotting predictions against actual values using scatter plots or line plots is another way to cross-check the SVR model. This will highlight any patterns or systematic bias.

In the event of poor performance, parameter adjustments like kernel type, epsilon, or regularization factor (C)may be required to improve generalization. When testing, cross-validation is frequently carried out to further validate the model's dependability.While neural networks undergo similar testing, they might also be more susceptible to the quantity and caliber of test data.Because NNs are capable of modeling intricate patterns, they are more likely to become overfit, which occurs when a model performs remarkably well on training data but poorly on fresh data.

In order to determine whether the neural networks have simply recalled the training data, their accuracy and loss are monitored during training on the test set.
Understanding the models' performance is aided by visualization tools like regression plots, learning curves, and confusion matrices (if classification is being done).
In the event that test performance is subpar, network architecture may need to be changed, regularization might have to start, or more information might have to be gathered. Both SVR and NN undergo rigorous testing and inspection to ensure that, when applied to actual data, the model is not only accurate but also sound and dependable.

3.1.5 Prediction in real time

Implementing a machine learning model for real-time prediction comes next after it has been trained and refined. Real-time prediction means that the model accepts new input data in most instances from a user, sensor, or real-time data stream and produces an

instantaneous prediction without interruption. Depending on the level of complexity and speed required, Support Vector Regression (SVR) or Neural Networks (NN) can be used for this.

To enable real-time use, the trained model is typically serialized and loaded into an operational system, like a web server, or it is loaded into a mobile application, where it receives continuously updated data and produces outputs in real time. When prompt and relatively simple decisions must be made, the SVR real-time prediction system typically performs well. Since SVR models don't need a lot of processing power to make predictions (after training), they can be used for tasks like demand forecasting, stock price forecasting, or even sensor-based systems where latency is an issue.

Before being fed into the SVR model for prediction, the input data is preprocessed, scaled, and reshaped as needed on the spot. To guarantee accuracy and consistency, the real-time data must undergo exactly the same preprocessing procedures as the training data. When neural networks are used for real-time prediction, this stage can be applied to far more complex tasks like voice processing, image recognition, or intelligent automation systems.

Although neural networks require more computing power, real-time NN models can now function flawlessly on embedded and mobile platforms thanks to the introduction of platforms like TensorFlow Lite, ONNX, and PyTorch Mobile.

When deployed in real-time, the NN model gets input through an interface (like a sensor, web form, or API) and generates predictions nearly instantly. In real-time systems, maintaining low latency, high precision, and robust handling of edge cases is essential, particularly in mission-critical applications like autonomous cars or medical devices.

3.2 SVR Algorithm

Support Vector Regression (SVR) is a machine learning algorithm of the Support Vector Machine (SVM) family. SVM is used primarily for classification tasks, but its variation SVR is used for regression tasks predicting a continuous outcome variable rather than class labels. The principle of SVR is essentially to find a function that predicts the target values with the highest precision and also keep the model as simple as possible with some allowance for error (epsilon).In contrast to the traditional regression methods that try to minimize the error of prediction for each point SVR attempts to fit the closest line

possible in a given constraint. That is, SVR does its best to make predicted values as close as possible to the actual ones but only penalizes those predictions that exceed this range of epsilon. This method allows the model to consume some amount of error without responding overzealously to data noise. SVR introduces the concept of an epsilon-insensitive tube a region of tolerance around the estimated function. Any data point that falls within this tube is considered an acceptable prediction, and no penalty is paid. But if a prediction falls outside the tube, a penalty that is proportional to the margin distance is incurred. The concept is to balance generalization with accuracy, so the model does not attempt to fit each point exactly (which would result in overfitting). The optimization objective of SVR is to discover a function f(x)=wx + b which deviates from the true target values by no more than ε for all the training data and is keeping w (the weight vector) as small as possible. Mathematically, it optimizes the cost function represented by equation.1:

$$\frac{1}{2}\|w\|^2 + c\Sigma(\xi_i + \xi_i^*) \quad (1)$$

Here, ξi and ξi* are slack variables that allow for errors beyond the epsilon margin, and **C** is the regularization parameter that controls the trade-off between flatness of the function and the amount up to which deviations larger than ε are tolerated.

- The slack variables measure how much a data point is away from being in the epsilon tube.
- When a point is inside the margin, the slack is zero.
- When it's outside, the slack captures the excess.
- The SVR model allows some slack so that it can still generalize well, especially when data are noisy.
- Such slack variables are necessary to prevent the model from being too strict
- Like SVM, SVR can impose
- the kernel trick to address non-linear relationships between data.
- The technique achieves this by projecting the input space into a space of higher dimensions where a linear regression line
- can be utilized.

- Common kernels include the Linear, Polynomial, and Radial Basis Function (RBF) kernels.
- The RBF kernel is widely applied because it can model complex, non-linear structures effectively.
- SVR has a number of hyperparameters that have a significant impact on its performance • C (Regularization
- Parameter) Controls the trade-off between a low error and a simple regression curve.
- Large C tries to fit the training data closely, while small C is simplicity-focused.
- Epsilon (ε): Specifies the size of the margin in which no penalty is assigned. Larger epsilon disregards more deviations and creates a simpler but less accurate model.
- Kernel parameters: e.g., gamma in the RBF kernel sets how far away from the centre one point has influence.

Support vectors are the points on or outside the epsilon tube that SVR finds during training. Since they determine the regression line's orientation and location, these are the most crucial points. No point inside the tube adds anything to the finished model.

The sparse solution from SVR works efficiently to avoid overfitting and provides faster computation. The important support vectors become the main focus of an efficient solution. SVR handles high the dimensional complexity of data poses no problems to SVR while it serves both linear and non-linear regression assessments. SVR applies to data situations with noise or outliers extremely effectively. With large datasets, SVR utilizes high amounts of computing power primarily because it works with complex kernel functions.

The implementation of SVR requires optimization of kernel C and epsilon and The kernel hyperparameters expensive and might need cross-validation. Time series analysis, environmental modeling, Multiple applications like electricity load forecasting and stock price prediction belong to the list of real-world applications of SVR. The system delivers successful results in situations which accept minute forecasting mistakes while rejecting major deviations. Models with large deviations are not possible due to the flexibility of subtle pattern recognition in SVR systems subtle fluctuations. The ability to strike an optimal equilibrium between flexibility and accuracy makes SVR

maintain its status as an important machine learning tool important tool in the machine learning arsenal.

3.3 NN Algorithm

A neural network functions by establishing a layered "neural" network that processes information identically to brain processing. Biological neurons create signal transmissions through this specific procedure. Neural Networks are particularly effective the state-of-the-art capabilities of these systems to uncover patterns and link data relationships make them appropriate for numerous applications.

The designed system reaches its best performance when used for both visual detection systems alongside voice recognition and time-based sequence examination requirements. A basic neural network consists of three distinct layers which start with an input layer that ends with hidden layers before finishing with an output layer network. Through mathematical operations the hidden layers recognize patterns while the input layer functions independently for pattern recognition. The neural network computation uses raw data features which it receives from each individual piece of information. The output layer functions as the system which gives prediction results through class labelling (classification) or a continuous value (for regression) to the prediction Each neuron in each the signal moves between neurons that belong to different layers by using weight values which control signal intensity the signal's amplitude.Forward propagation is the first step in a neural network's learning algorithm.

Layer by layer, all of the input features are transmitted throughout the network, and mathematical operations like activation functions and weighted sums are computed. These computations change the information and permit non-linear relationships to be learned by the network. Finally, a loss function is used to match the output with the true label.

In order for the neural network to learn complex problems, activation functions add non-linearity. Activation functions used commonly are ReLU (Rectified Linear Unit), which returns zero for negatives and the same number for positives, Sigmoid, which scales outputs between 0 and 1, and Tanh, which scales between -1 and 1. The activation function influences the learning ability of the network during training. The network uses a loss function to calculate how far the prediction deviated from the actual outcome.

Mean Squared Error (MSE) and Mean Absolute Error (MAE) are common loss functions for regression tasks. Cross Entropy Loss is used more frequently for classification tasks. The more accurate the network, the lower the loss. The network adjusts its internal weights to minimize the error after the loss has been calculated. Backpropagation, in which the error is transmitted from the output layer to the input layer in reverse, is how this is accomplished.

Then, using gradient descent, an optimization technique that iteratively learns the network on its errors, each weight is updated based on how much of the error it is responsible for.

Updates on Gradient Descent the weights by calculating the loss function's gradient, or slope, in relation to each weight. To reduce the error, the algorithm moves in the direction of the negative gradient. By employing momentum or dynamically modifying the learning rate, some variants, like Adam, RMS Prop, and Stochastic Gradient Descent (SGD), improve learning speed and efficiency. Many input-output pairs are used to train a neural network.

At various epochs (runt through all the data), the model next learns to change its weights step by step in an attempt to do better. It is necessary to take precautions against overfitting, which occurs when a model performs well on the training set but poorly on fresh, untested data. This can be prevented by using strategies like early stopping, regularization, and dropout. A few types of neural networks are specific to a particular task.

1.The most basic kind, feedforward neural networks (FNNs), only allow data to flow in one direction.

2.Convolutional Neural Networks (CNNs) are used to process images and videos.

3.Recurrent Neural Networks (RNNs) are excellent for sequential data, such as natural language or time series.

4.To learn intricate patterns, Deep Neural Networks (DNNs) use a large number of hidden layers. There are many real-world applications for neural networks.

They enable voice assistants, suggest products on shopping websites, detect diseases in medical images, and even enable self-driving cars. Their utility is that they can learn from

data without being programmed with rules explicitly. With advancing computing power and data availability, NNs keep getting more powerful and expanding the limits of artificial intelligence.

CHAPTER 4
RESULTS AND DISCUSSION

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The simulation platform is intended to examine and optimize beamforming methods for next generation 6G wireless systems based on Artificial Intelligence (AI). Python 3 is chosen as the primary simulation platform, using strong deep learning libraries like Keras and TensorFlow for model development, training, and testing. The simulation environment is in urban outdoor in an area of interest focused on a high-frequency band of 28 MHz atypical millimeter wave frequency used in next-generation wireless communication systems. The configuration is single base station with a Uniform Linear Array (ULA) of 32 antenna elements. On the user side, the simulation supports 100 user equipment (UE) devices, one with a single isotropic half-wave dipole antenna. These UE antennas are rotated randomly by 0°, 30°, 60°, and 90° to simulate actual variability in user orientation. Three commonly employed beamforming algorithms are considered in this research: Maximum Ratio Combining (MRC), Zero-Forcing (ZF), and Minimum Mean Square Error (MMSE).

The algorithms are compared according to their purpose, interference handling capacity, impact on noise amplification, complexity, and performance. MRC maximizes the signal-to noise ratio (SNR) by implicitly weighting received signals and is characterized by low complexity and satisfactory performance in non-interference-prone cases. ZF, however, directly sets interference to zero by essentially removing inter-user interference but at the expense of higher noise amplification and moderate complexity. MMSE attempts to find a compromise by setting the mean squared error to a minimum with consideration to both noise and interference. It usually provides optimal performance but is also computationally the most demanding.

Based on these trade-offs, Maximum Ratio Combining (MRC) is chosen for the simulation because it is simple and has a good baseline performance, particularly in cases where there is little interference. For data preparation, the DeepMIMO dataset generator is used. The generator uses REMCOM Insite's 3D ray tracing engine to create an extremely realistic wireless propagation environment. The chosen scenario, "O1_28," is a preconfigured outdoor urban environment optimized for the 28 MHz frequency band and optimized to mimic data for 100 users. DeepMIMO supports high customization. For this simulation, some parameters are set, for example, specifying the

scene between 1 and 5, restricting the number of propagation paths to 10, and specifying active base stations and users within certain rows. Data sampling is established with row and user subsampling at 0.5 to have a balanced yet diverse set of users.

The base station antenna is considered a ULA with 1 row, 32 elements, and 1 column, where the element spacing is half-wavelength. It employs a half-wave dipole radiation pattern and rotates by 0°, 30°, and 90° and provides directional diversity in signal propagation. Likewise, every user terminal is provided with an isotropic antenna and implemented with various rotations like (0,30), (30,60), and (60,90) degrees to represent realistic dynamics. Other parameters are to facilitate base station-to-base station (BS2BS) communication and set up Orthogonal Frequency-Division Multiplexing (OFDM). The parameters in OFDM determine 512 subcarriers, limited to 64 subcarriers for computational purposes, a sampling rate of 1, and a bandwidth of 0.05. All these settings together enable high-throughput data transfer, frequency selectivity, and robust multi-user communication analysis.

Finally, the simulation setup and dataset preparation have been carefully established to develop a realistic test bed to benchmark beamforming methods. The employment of the physical layer fidelity of simulations stems from DeepMIMO because it functions as an advanced 3D ray tracing data simulator. The system enables users to make directed selections of beamforming protocols combined with antenna configuration while maintaining physical layer accuracy depth performance evaluation of MRC in realistic 6G implementations.

4.1 Frontend Output

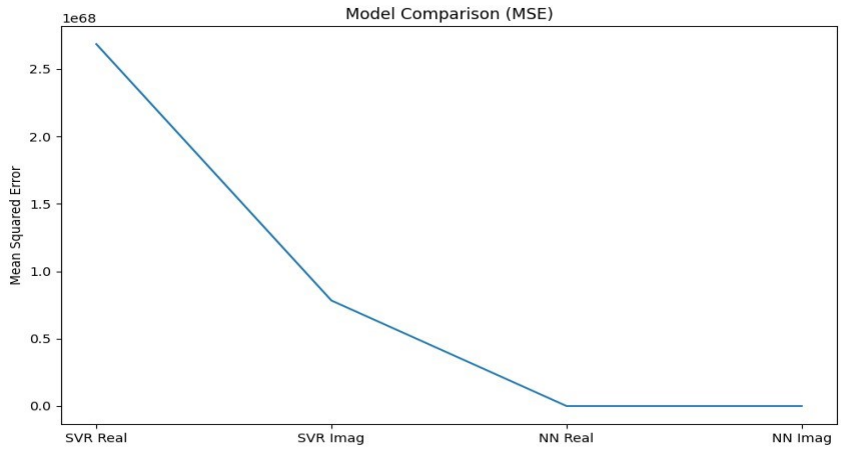


Fig 4.1: Model Comparison plot

The examination of prediction accuracy relies on Mean Squared Error (MSE) values which are used in "Model Comparison (MSE)". The MSE metric functions as the primary evaluation basis in the "Model Comparison (MSE)" graph for assessing model effectiveness models. The diagram depicts the four separate model elements with SVR Real at the left and SVR Imag to its right followed by NN Real and NN Imag at the right. The y-axis in the graph displays MSE values from 1068 to represent the evaluation metric as the x-axis contains the four distinct model components—SVR Real, SVR Imag, NN Real, and NN Imag a logarithmic scale with extremely high values, especially for some models.The charts'

The downward direction of both lines illustrates that SVR models demonstrate inferior performance compared to NN models according to "Model Comparison (MSE)." The NN (Neural Network) models demonstrate better prediction accuracy than SVR (Support Vector Regression) models do. The SVR Real model displays an exceptionally high MSE value above 25 × 1068 that demonstrates its limited capability. The data prediction capability of SVR models is weak for real values. The performance signals that its possible limitations stem from the model itself. Support Vector Regression models exhibit limited capability to handle non-linear patterns and therefore performs worse than deep learning models in data capture the underlying complexity or variability in the dataset's real part. In contrast, SVR Imag delivers better outcomes for predicting imaginary parts despite running on a dataset with a corresponding high error rate. The system produces predictions with an average error rate that remains high. An improved performance can be gained through

The modified NN Real model produced near zero MSE in determining results. The neural network demonstrates an excellent capability to understand the real component of the data. The data modeling ability of multiple hierarchical abstractions enables the system to identify complex patterns probably explaining its success levels of abstraction. With the same small MSE, the NN Imag model performs just as well, demonstrating neural networks' capacity to handle both real and imaginary data components in this scenario.

In terms of prediction accuracy, this graph unequivocally shows that neural networks perform noticeably better than support vector regression models, particularly in applications involving complex-valued data such as beamforming weights in communication systems. Deep learning techniques are effective for tasks requiring high

precision, as evidenced by the sharp decrease in MSE from SVR to NN. Designing AI/ML models for Digital Twin environments in 6G systems, where prediction accuracy can directly affect real-time beamforming, channel modeling, and signal optimization, benefits greatly from such insights.

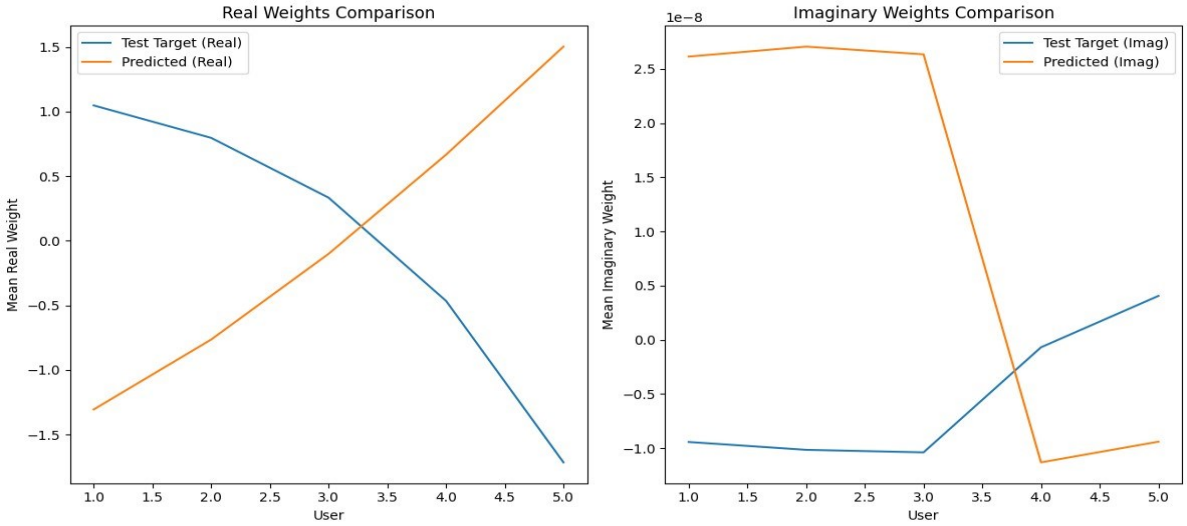


Fig 4.2: NN weights comparison

Two line graphs comparing the actual (test target) and predicted beamforming weights real and imaginary components across five users are included in this image. The Real Weights Comparison is depicted in the graph on the left, and the Imaginary Weights Comparison is shown in the graph on the right. When using digital twin and AI/ML models for applications like signal steering and directional communication in 6G systems, these plots are essential for assessing how well a model has learned to predict the beamforming weights.

The test target values (blue line) in the Real Weights Comparison graph show a steady decline from approximately +1.1 to -1.6 as the user index rises from 1 to 5. On the other hand, the expected values (orange line) show an upward trend, beginning at -1.4 and reaching roughly +1.5A stark divergence between the real and expected actual weights is presented by this opposite trend. The model has failed to capture the underlying distribution well or has overfitted to different features of data if it cannot determine the correct pattern or trend. Test target and predicted values on the Imaginary Weights Comparison chart are both near the 10-8 scale, implying that imaginary parts are extremely small likely to be insignificant in this instance.

For users 1 to 3, the forecasted values are positive and practically level; however, at user 4, the suddenly fall to negative values. Despite modest overall values, this sudden transition, especially in the case of users 3 and 4, suggests that the models prediction for imaginary parts is not well agreeing with the test targets. Generally speaking, the real weights are critically significant, and the models prediction is far off from the target, even though the imaginary weights scale can imply that they play less a role in the model. In 6G beamforming, where weight prediction accuracy determines the effectiveness of signal direction and quality this difference can be critical. From the analysis, the model needs to be fine-tuned or retrained, perhaps with deeper network structures or improved feature engineering, in an attempt to capture the ideal beamforming weights and user characteristics more effectively.

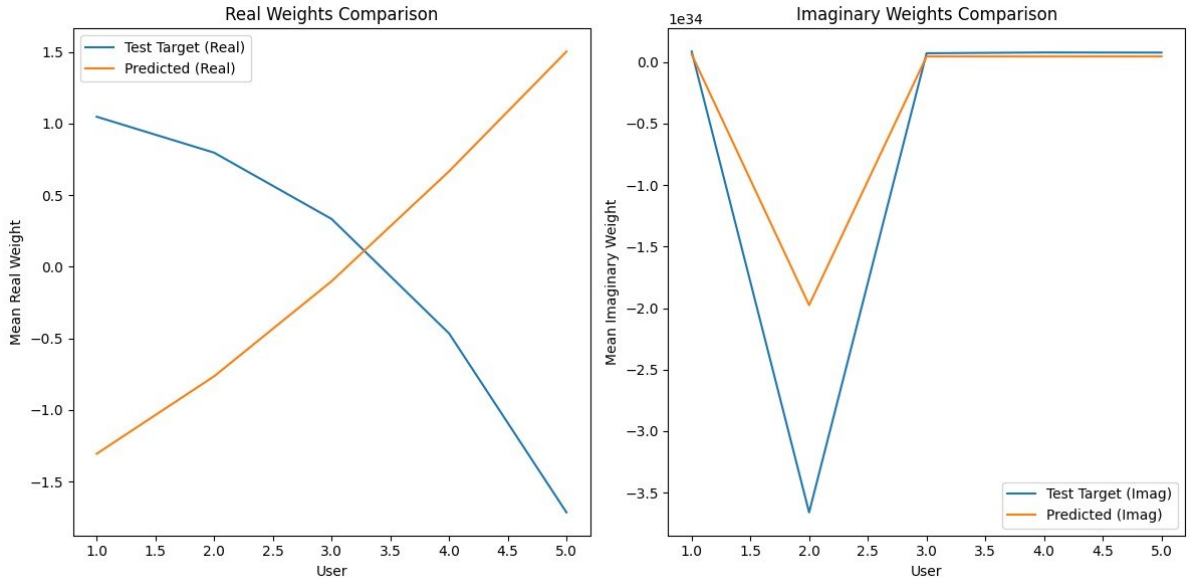


Fig 4.3: SVR weights comparison

The two comparative plots for beamforming weight predictions, the left graph for real components and the right graph for imaginary components, are examined across five users in this updated image. The diagrams reveal that the prediction model delineates the appropriate beamforming patterns effectively. The model demonstrates its ability to replicate proper patterns through a comparison between expected values (orange) and actual test targets (blue lines). Testing precision through model-static target analysis allows us to check the accuracy levels of AI/ML-based digital twins.

The application of AI/ML achieves better results for wireless systems especially those advancing to 6G technology with beamforming requirements precision is crucial. The

Real Weights Comparison plot demonstrates how real target weights change when the user index raises between 1 to 5. Real target weights in the left panel maintain a continuous downward trajectory starting from +1.1 through to -1.6 about -1.6. The linear trend of expected values moves from about -1.4 to +1.5 during this assessment. A substantial trend discrepancy between the prediction and ground truth is indicated by this opposing direction. The performance of directional signal transmission in real-time scenarios may be impacted by such a discrepancy, which suggests that the model is unable to learn or capture the inverse relationship shown in the real component. This could result in high mean squared errors.

Potential numerical instability or anomalous behaviour in the dataset or the prediction model is highlighted by the Imaginary Weights Comparison plot on the right, which shows data on a massive scale roughly 10^{34}. At User 2, both the test target and predicted curves show a significant dip, with values falling below -2×10^{34} for the actual and marginally higher for the prediction. Following this, both values for the other users abruptly fall back to near zero levels. This indicates a sudden, unexplainable spike that may be due to overfitting, incorrect normalization, or loss of accuracy, or maybe a singularity in the models calculation that drove the weights to extreme values. These plots together indicate a serious fault in the existing modeling method. The imaginary component indicates instability or potential issues with the data or learning process, while the real component always shows error due to inverted trends. It may be worthwhile to pursue actions such as rerunning input feature distributions, applying normalization methods, data diversity in training, or model architecture alteration to enhance model dependability. Those changes are required in beamforming applications powered by digital twin technology to ensure correct and consistent modeling of real and imaginary parts.

CHAPTER 5
CONCLUSION AND FUTURE WORK

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5.1 Conclusion

In this project, we were able to develop an end-to-end simulation environment to test beamforming methods in a 6G communication environment through AI-based software. With the use of Python 3 and TensorFlow and Keras, in conjunction with the REMCOM in sites 3D ray tracing-powered DeepMIMO dataset generator, we created an outdoor real-world urban setting with high-frequency signal propagation at 28 MHzThe simulation scene included a Uniform Linear Array of 32 antennas and a base station, with 100 different user devices having varying orientations of antennas. This gave us the opportunity to validate the system performance in conditions that best reflect real-world 6G deployment scenarios. Among the considered beamforming techniques Maximum Ratio Combining (MRC), Zero-Forcing (ZF), and Minimum Mean Square Error (MMSE) MRC was used in this work because it has low complexity and can efficiently maximize signal-to-noise ratio (SNR) in low-interference conditions. While ZF and MMSE provided good interference suppression and accuracy, they have higher computational costs, which may not be feasible in systems where power and processing capabilities are limited.

The implementation of MRC successfully maintained a suitable connection between system performance and its simplicity. The basic design of the approach made it function as both an evaluation starting point and an execution framework thus earning recognition in the field. The process several simulation evidence sources depend on the generation process for the DeepMIMO dataset to establish simulation credibility. With this system users can precisely manage antenna positioning as well as user position and wireless signal paths. Through the method users obtained exact result duplications while conducting experiments with various alternative parameters user dynamics and antenna behaviours under various orientations. The addition of OFDM **t**he analysis combined various OFDM parameters to obtain complete signal quality evaluation from different subcarrier frequencies. The wireless systems reach their high operational speed through this vital operational factor. Overall, this research uses artificial intelligence together with realistic dataset generation as its primary example. Different researchers join forces to analyze complex wireless communication systems through their studies beamforming. The evaluation of MRC performance happened through a virtual

simulation environment. The simulated modeling provides authentic data about successful and unsuitable elements of this approach within a 6G environment. Research needs to focus on shifting user conditions to develop analysis methods that deliver optimal results for investigators. Performance enhancement depends on creating advanced antenna systems together with machine learning methodology for beamforming operations. The research creates base knowledge for upcoming investigations to advance their research. Wireless communication systems driven by AI constitute the wireless technology of upcoming generations.

5.2 Future Work

The research investigated Maximum Ratio Combining (MRC) through practical application for 6G beamforming under this project. Research opportunities within 6G scenarios exist extensively regarding the application and investigation of Maximum Ratio Combining (MRC) for beamforming implementation. The development of beamforming methods can continue through implementation tests on Zero-Forcing (ZF) combined with other variations and Minimum Mean Square Error (MMSE) in real-world applications. The algorithms are the system functions at various levels to compare performance metrics and complexities and noise amplification and interference handling capability interference handling ability, and application.

By implicitly weighting received signals, MRC achieves the highest signal-to-noise ratio (SNR). It stands out for its low complexity and respectable performance in situations where interference is not a problem. By effectively removing inter-user interference, ZF, on the other hand, explicitly sets interference to zero at the expense of moderate complexity and increased noise amplification. By minimizing the mean squared error for both noise and interference, MMSE aims for equilibrium.

Although it is the most computationally demanding, it usually provides the best performance. Maximum Ratio Combining (MRC) is chosen for simulation based on these compromises because it is simple and performs well at baseline, particularly in situations where there is not much interference. The DeepMIMO dataset generator is used to prepare the data.

The generator simulates a highly realistic wireless propagation environment using the REMCOM Insites 3D ray tracing engine. O1_28, the scenario that was chosen, is a

preconfigured outdoor urban setting that is optimized for the 28 MHz frequency band and designed to replicate data for 100 users.

Lastly, the future can include hardware-in-the-loop testing, where software models are combined with actual hardware such as software-defined radios (SDRs) in order to test beamforming algorithms in the physical world. This would close the gap between simulation and real-world deployment. It would also be interesting to compare how the system performed with different propagation models and above-frequency bands, such as terahertz bands being discussed for future 6G generation standards. Finally, the future can involve hardware-in-the loop testing, whereby software models are merged with hardware like software-defined radios (SDRs) in an effort to test out beamforming algorithms in the real world. This would bridge the gap between simulation and real-world deployment. It would also be interesting to study the energy efficiency and latency of beamforming methods under AI management, as they are significant parameters for 6G applications like autonomous vehicles, smart cities, and IoT networks. All these future contributions together will play a crucial role in refining the simulation into a more stable, scalable, and deployment-ready framework for next generation wireless systems. Another area of future work of interest is the integration of real time CSI estimation into the beamforming process CSI in actual wireless channels is critical for beam direction estimation and gain.AI models like CNNs or RNNs can efficiently predict or estimate CSI, and if performed with high accuracy, can significantly enhance the adaptability and fault tolerance of beamforming in highly dynamic urban environments. Developing light-weight models for CSI prediction that can be deployed on edge devices will further facilitate real-time deployment in systems with limited resources like mobile phones or IoT devices. Furthermore, there is scope for exploring multi-user beamforming and user clustering methods. In areas with high population density in cities, a base station must support many users simultaneously, typically with overlapping channels. Advanced methods like coordinated beamforming, NOMA (Non Orthogonal Multiple Access), or joint transmission in MIMO systems can be employed for increasing spectral efficiency. Machine learning algorithms can assist in grouping dynamic users with similar spatial properties, thereby maximizing the network's overall throughput and minimizing interference. These techniques will be especially relevant as user traffic loads grow in 6G-smart cities and autonomous systems.

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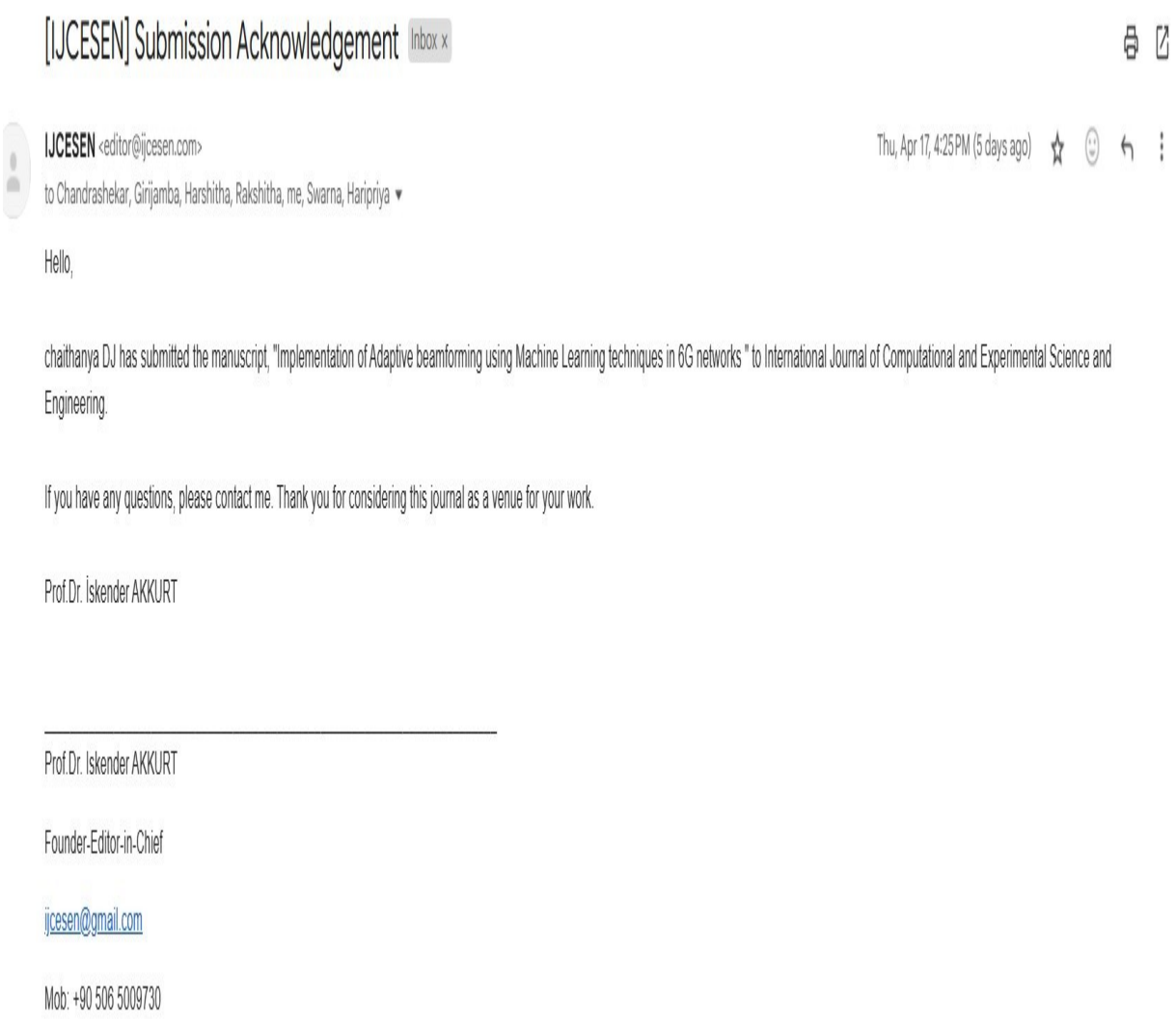
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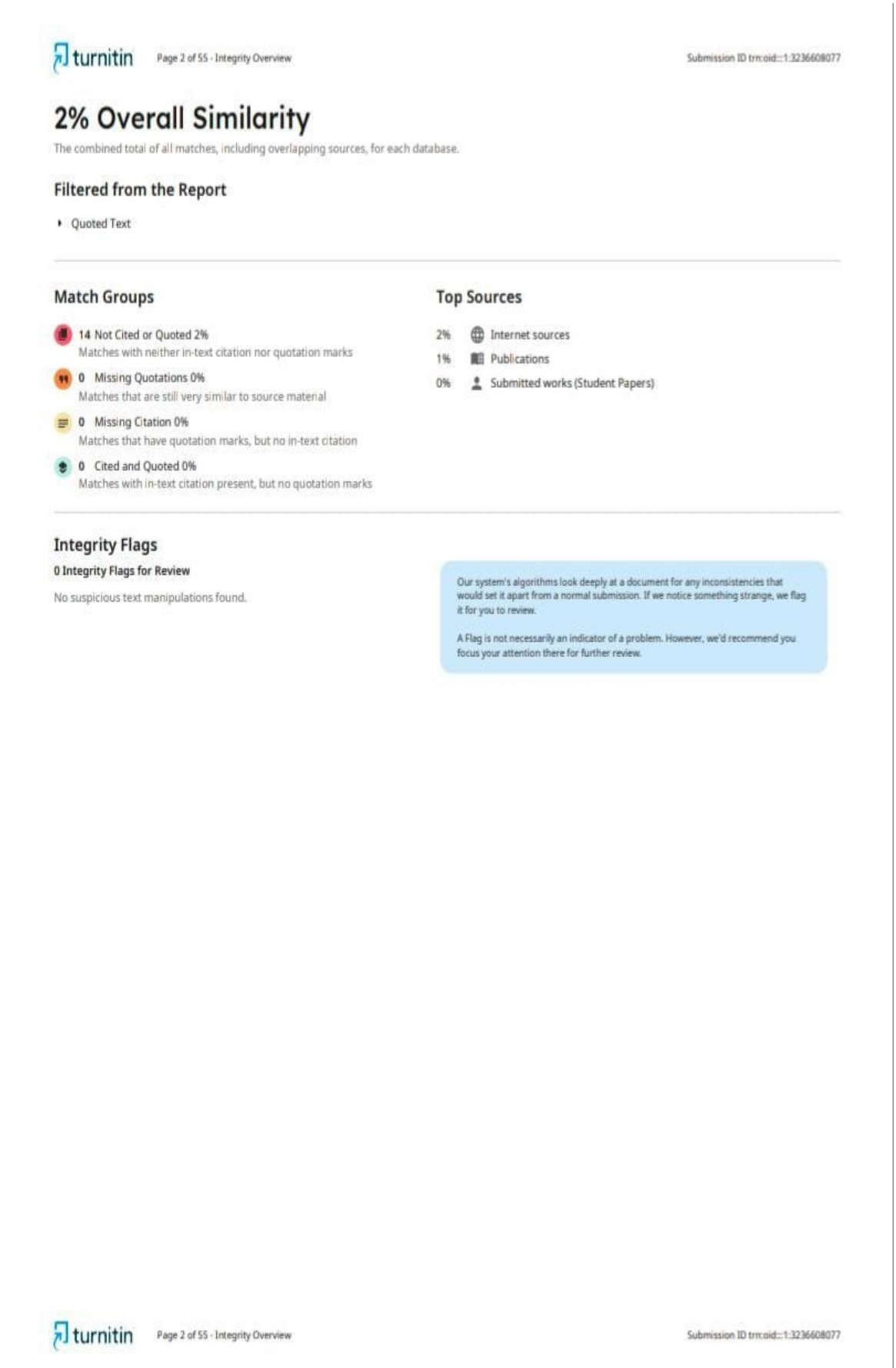
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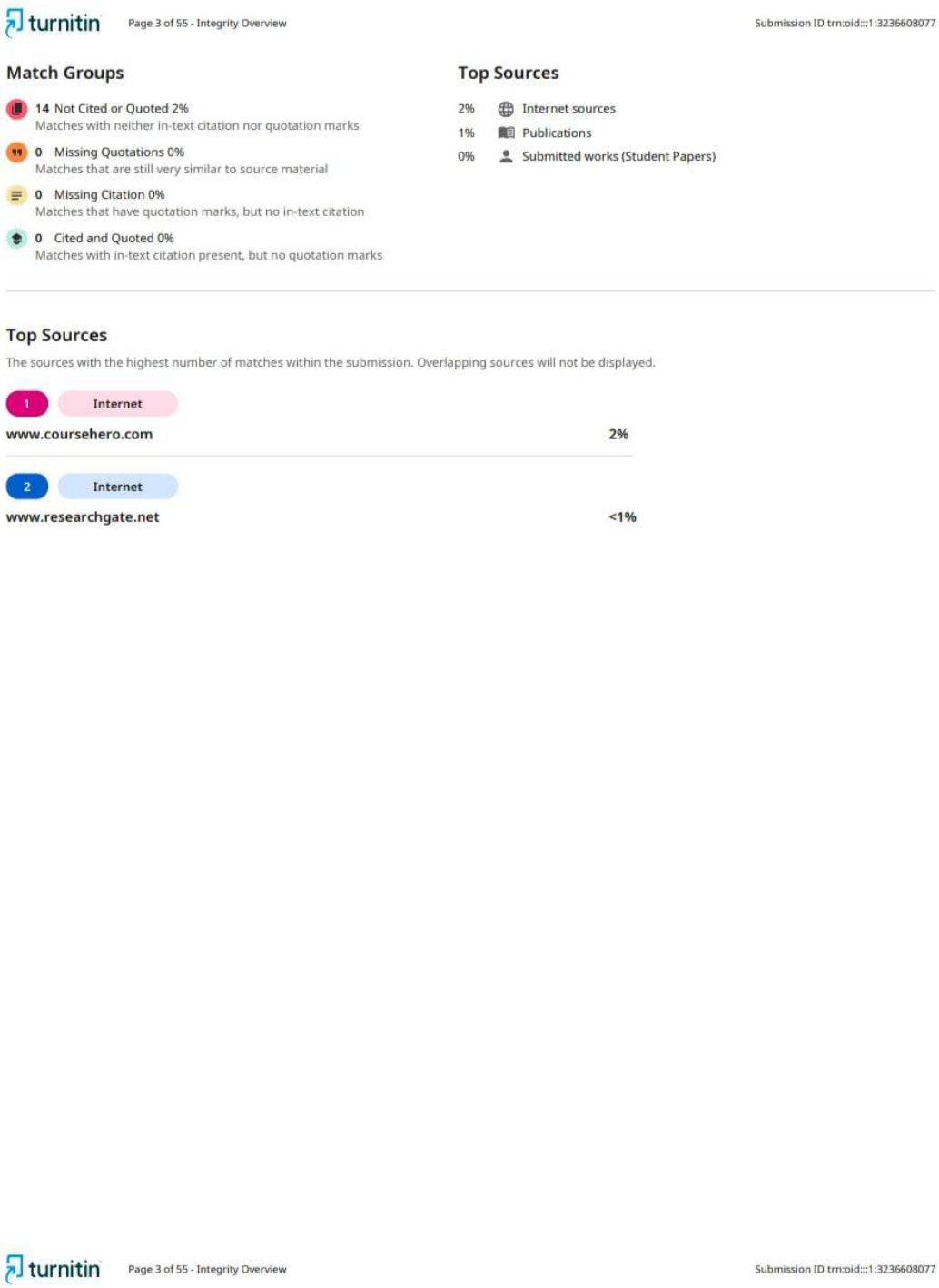
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Implementation of Adaptive beamforming using Machine Learning techniques in 6G networks

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Abstract:

In modern wireless networks, especially within the evolving landscape of 5G and the anticipated 6G systems, efficient parameter optimization is essential. While classical optimization techniques focus on enhancing operational efficiency, machine learning (ML) introduces predictive intelligence, both of which are critical for next-generation network performance. This paper proposes the integration of ray tracing with intelligent beamforming strategies to address the limitations of conventional techniques in dynamic environments. Traditional beamforming techniques cannot be adapted to dynamic communication environments. Based on neural networks and SVR, our proposed model will implement real-world channel data to optimize signal transmission and reception. Beamforming, a key enabler for improved spectral efficiency in 6G, requires adaptive and precise signal direction. To achieve this, the proposed approach leverages artificial intelligence and machine learning (AI/ML) techniques, specifically artificial neural networks (ANNs) and support vector regression (SVR) with 32 antennas and 64 parameters, to enhance beamforming decisions in 5G New Radio systems. The results demonstrate that the AI/ML-based model effectively optimizes the beamforming process, offering a robust and scalable solution for improving the overall performance of future wireless 6G communication systems.

1. Introduction

The fifth generation's high success rate brings optimistic expectations to the next-generation 6G network, facilitated through key performance indicators such as ultra-reliable low-latency communication (URLLC), extremely high data rates, high energy efficiency, high spectral efficiency, ultra-dense connectivity, integrated sensing and communication, and secure communication features [1]. Beamforming serves as a vital signal processing method that powers wireless communication systems throughout 5G networks and future emerging 6G networks. It allows antennas to direct the transmission or reception of signals toward a specific direction rather than broadcasting them in all directions. This results in stronger, more efficient, and interference-free communications. With the increase in the number of users on the Internet of Things (IoT) environment, 6G communication systems have attracted the attention of researchers worldwide to provide 6G networks that can support many users with up to gigabit-per-second (Gbps) data rates in the millimetre-wave domain in consideration of big-data (BD).

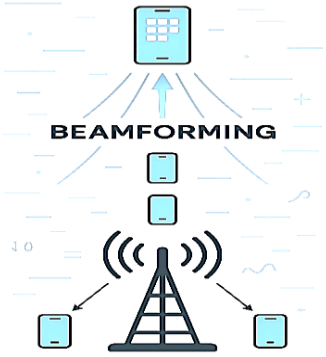


Figure 1. *The Beamforming*

The 5G millimeter-wave (mm-wave) communication system will be the place of seamless connection between several billion user devices and low latency, thus offering gigabit per second (Gbps) speed in a 5G environment, owing to the mm-wave signal transmission and secure connection [2]. Beamforming is a technique that directs signals that are transmitted towards specific directions instead of spreading the signals over a wide data area, as shown in Fig.1. This targeted approach improves the signal quality and reduces interference, which is a key enabler of high data rates and low latency for wireless communications [3]. Traditional beamforming operates in the far-field region, where the transmitted signals form planar wavefronts, and the energy is steered toward a general direction. However, with the rise of massive multiple-input multiple-output (MIMO) and extra-large (XL) antenna arrays, near-field beamforming can be performed. In the near-field, wavefronts are no longer planar but spherical, and instead of steering, the system can focus the beam precisely at a specific location, such as a mobile user device. This technique is known as near-field beam focusing and enables high-resolution spatial targeting, leading to better signal quality, capacity, and security in high-density or edge environments, as shown in Fig.2.

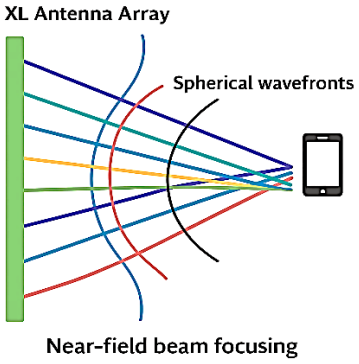


Figure 2. *Near-field beam focusing*

From the existing survey, two models are proposed in this work

- Neural networks are sequential or exhibit multiple features over time. Because we propose 32 antennas and 64 parameters, this method works well.
- SVR (Support Vector Regression) is a simpler model for predicting numerical values

The main contribution of this work is:

- AI-driven Optimization: Leveraging a pre-trained AI model to determine the optimal beamforming weights, bypassing manual calculations.
- Efficient Signal Processing: Transmitting the user's signal to a nearby server or cloud infrastructure for real-time processing.
- Instantaneous performance enhancement: Using the learned experience of the AI model to rapidly calculate and apply the optimal beamforming weights improves system efficiency

Section 1 introduces this study. Section 2 presents a survey and its outcomes. The main architecture of 6G is explained in Section 3. Section 4 discusses the methods and their implications. Section 5 presents and discusses the results. Finally, Section 6 concludes the study and outlines future work.

2. Related works

The fifth generation (5G) wireless communication system selects millimeter-wave (mm-wave) frequencies because these frequencies offer wide-spectrum resources. The wide availability of bandwidth in wireless communications does not

mitigate path loss during operation at millimeter-wave frequencies. Beamforming within the massive MIMO technology enables the detection of path losses with increased effectiveness [4]**.** The author discusses incorporating artificial intelligence (AI) and machine learning (ML) into 5G wireless standards can lead to networks that can operate autonomously and reconfigure themselves with very low latency. This implies that the network can quickly adapt to changing conditions without human intervention [5]. The evaluated adaptive beamforming works within fifth-generation massive multiple-input multiple-output millimeter-wave cellular networks. The system automatically forms dynamic beams based on traffic requirements to enhance spectral and energy efficiency with low computational and protocol complexity [6].

Beamforming techniques in the context of mm-wave and bandwidth handling offer a way for wireless networks to deal with the vast bandwidths that are to be supported in the future. The implications of this hold for advancing wireless technology and the potential for creating faster, more efficient networks are endless. [7]. The 3rd Generation Partnership Project (3GPP), fifth generation (5G)-advanced new ratio (NR) standardization, has begun research on artificial intelligence (AI) and machine learning (ML)-based beam prediction systems for future commercialization as part of their standard evolution to sixth generation (6G) communications [8].

In the context of 5G new radio beamforming, the author proposes a novel application of artificial neural networks. This approach has been shown to successfully optimize a very complex mechanism of ANNs through its sophisticated mechanisms to optimize beamforming as a process and improve the overall performance of the 5G new radio system. The use of ANNs enables real-time adaptation and decision-making, thus overcoming the deficiencies of traditional beamforming techniques [9].

Performance of a deep learning approach deployed on fifth generation (5G) millimeter-wave (mm-wave) multicellular networks. For this purpose, two NNs are properly trained to the optimum beamforming configuration, which is defined in terms of mean square error (MSE) minimization. The first network takes the input of the required spectral efficiency (SE) per active sector, whereas the second network has the input of the corresponding energy [10].Performance evaluation of base station adaptive beamforming methods is integrating massive multiple input multiple output (MIMO) in all active base stations, within a fifth-generation millimeter-wave multicellular network. Different beamforming configurations can be applied to meet special requirements, although predefined sets operate for the requested directional beam steering during traffic management. The k-nearest neighbor (k-NN) approximation of machine learning (ML) serves as the beamforming method for creating suitable configurations by extracting data from throughput demand distributions within the network area [11].

The author in his study has discussed deep-learning-based beamforming for signal processing to enhance wireless communication systems. Traditional beamforming techniques cannot be adapted to dynamic communication environments. Based on convolutional neural networks (CNNs), our proposed model learns spatial features from real-world channel data to optimize signal transmission and reception. To simulate realistic communication scenarios, the author used a synthetically generated indoor mm-wave channel dataset that included multiple moving targets [12].

The development of 5G technology systems since the 5G era indicates that adaptive beamforming and DSP strategies can contribute to the design of 5G wireless-transformation capabilities. Phase-shift beamforming performance evaluation takes place in a system that uses Artificial Intelligence (AI) to improve link and communication quality in dense urban areas using a Uniform Rectangular Array (URA) system [13]. The author proposed a novel action space, in which the subset of the beam sweep to be covered can be adjusted based on high- and low-frequency channel propagation laws [14].To minimize the downlink beam sweep overhead of the mm-wave system, a deep reinforcement learning based dynamic beam selection (DRL-DBS) technique is proposed. The spatial analysis of high - and low-frequency channel laws serves as the basis of a new learning principle for identifying the spatial spectrum index difference between optimal mm-wave and sub-6 GHz beams. The proposed actions to change the size of the region of the beam sweep subset were used by the DRL-DBS method based on the high- and low-frequency channel propagation laws [15].

Reconfigurable intelligent surfaces (RIS) technology is a product of innovation that can transform the performance metrics of 6G wireless networks. The author covers the intricate dispersion of RIS technology through schematic frameworks and implementation barriers, along with expected future trends [16]. The author discusses cooperative ISAC in cell-free multiple-input multiple-output (MIMO) systems, where multiple MIMO access

points (APs) collaboratively provide communication services and perform multi-static sensing. Formulate an optimization problem for the ISAC beamforming design that maximizes the achievable sum-rate while guaranteeing the sensing signal-to-noise ratio (SNR) requirement and total power constraint [17].

The main goals of 3GPP-based standard studies are to minimize the overhead from reference signals (RSs) and reduce the number of beam sweeps at the user equipment (UE), which arise owing to frequent beam measurements caused by UE movement and rotation [18].The author explained two optimization methods and two classical reinforcement learning approaches to determine the optimal beamforming angles for the transmitter and receiver antennas [19].The author discusses hybrid MIMO systems that can improve communication efficiency by using fewer RF chains than antennas. This is particularly important for future networks, such as 6G, where the number of antennas can reach thousands. By optimizing the use of resources, hybrid MIMO systems can effectively manage the costs and power consumption [20].

Exploiting beamforming techniques to direct radio waves into narrow beams targeted at users is essential for 5G and 6G networks to ensure high data rate requirements with low latency. However, beamforming in a dynamic environment is challenging owing to its high sensitivity to shifts in receiver location. Therefore, an adaptive beamforming approach was required. The authors utilized Deep Sense 6G and 5G MIMO datasets for optimal beam prediction using deep learning approaches. A simple approach to transform the radar input into vector embeddings in the Deep-Sense 6G dataset. Moreover, the authors addressed the severe minority class problem in the 5G MIMO dataset [21].

The proposed approach addresses the specific challenges posed by millimeter-wave frequencies, train mobility, diverse propagation, and environmental conditions. The methodology integrates the collection of a comprehensive dataset that captures the environmental conditions and encompasses the different characteristics of the train movement. To ensure accurate and timely predictions of 3D beam pairs, they carefully developed and compared various multiclass supervised machine learning classification algorithms [22] [23].

This study does not specifically address the simulation of beamforming using Artificial Intelligence and Machine Learning (AIML) in 6 G. It focuses on asymmetric communication systems and the application of a space-time-frequency non-stationary channel model to analyze scattering distributions and statistical properties influenced by different antenna patterns. While beamforming is discussed, the integration of the AIML for simulation purposes is not covered in this research [24].

The author uses space-time-frequency non-stationary channels to model asymmetric channels through parameter generation with uplink-downlink correlation using diverse antenna arrangements [25].Emerging technologies such as intelligent reflecting surfaces (IRS), unmanned aerial vehicles (UAVs), and non-orthogonal multiple access (NOMA) can provide communications for massive users, high overhead, and computational complexity [26].

The main outcome of the survey is a data-driven neural network system that selects beams during transmitter-receiver pair operations. A beam selection method based on SVR and neural network uses a simulated beam dataset for training. Neural networks are powerful tools for supervised data mining tasks, particularly for categorical and continuous target variables. They mimic the structure and function of human brain neurons and excel in complex non-linear problems. However, the implementation of neural networks faces challenges until the development of algorithms, such as backpropagation. Neural networks require careful data preparation, model complexity management, and validation to ensure accurate prediction.

3. Architecture

6G wireless communication systems present new opportunities to leverage artificial intelligence (AI) and machine learning (ML) techniques to enable intelligent networking. This study examines the application of AI and ML in intelligent networking for 6G, aiming to enhance network performance, optimize resource management, and improve user experience. However, the application of AI and ML to 6G networks presents several significant challenges. The design and implementation of intelligent networking algorithms must address issues such as scalability, real-time decision making, energy efficiency, privacy, and the need for robust learning in dynamic and heterogeneous network environments [27]. Sixth generation (6G) mobile communication utilizes beamforming technology to dynamically adjust the beamwidth with beam direction control to maximize system performance. The all-digital beamforming method in asymmetric communication systems enables numerous user interactions in the uplink combined with a superior

SNR in the downlink. The modelling accuracy of channels is vital for asymmetric communication systems.

Connectivity establishment and maintenance of network radio nodes and User Equipment (UEs) require BM protocols as an essential link. Distributed multiple-input multiple-output (D-MIMO) systems comprise a Central Processing Unit that coordinates multiple Access Points used to serve multiple user equipment devices. Because of the numerous downlink (DL) reference signal transmissions required for beam sounding of millimeter-wave frequencies, it becomes difficult to select the optimal Access Point (AP) beam to serve User Equipment (UEs) [28] . Utilizing a software program called RCOM to model 5G signals would be practical. In the Urban Outdoor Terrain, users are in an urban outdoor environment, with buildings around, just as it would be in real life. The frequency band (28 MHz) is the rate at which 5G signals are transmitted. They are widely used in 5G, particularly for rapid data transmission.

Fig.3 shows the architecture and key enabling technologies of a 6G wireless communication network. The 6G Core Network serves as the backbone for high-speed, low-latency, and intelligent connectivity. 6G allows multiple virtual networks to operate on a shared physical infrastructure, each tailored to specific applications such as autonomous vehicles, smart healthcare, or industrial automation. Connected to this architecture is Massive Machine-Type Communication (mMTC), designed to support a vast number of IoT devices with minimal human intervention.

To improve the efficiency and adaptability of these systems, Artificial Intelligence (AI) is integrated to manage resources, predict network behaviour, and provide intelligent automation. The processing of data is closer to the user equipment (Smartphones, Tablets, IoT devices), resulting in reduced latency and, hence, a better real-time response, which is key for applications such as AR/VR and autonomous drones. Alternatively, intelligent surfaces or reconfigurable intelligent surfaces (RIS) can be used to control electromagnetic waves to enhance signal coverage and quality. Surfaces are crucial for beamforming to enhance wireless signal propagation in complex environments. In summary, this architecture demonstrates that 6G incorporates strong AI, edge computing, intelligent surfaces, and network slicing technologies to empower highly adaptive, resource efficient, and scalable communication systems for next-generation wireless networks.

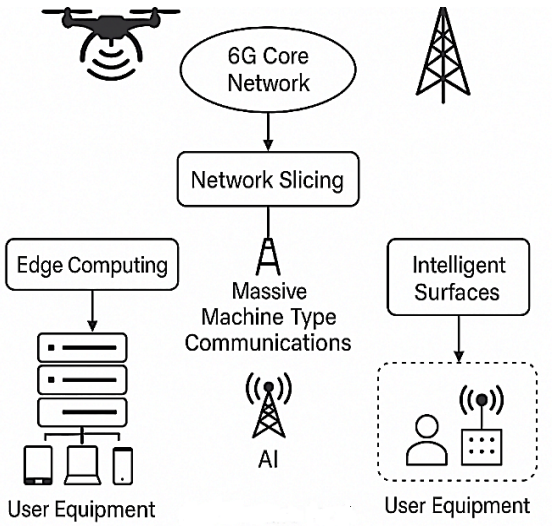


Figure 3. *6G Architecture*

The base Station utilised one base station (such as a tower-emitting signal). It contains a peculiar type of antenna called a Uniform Linear Array (ULA) with 32 antenna elements. This enables the base station to emit thin beams and transmit signals precisely in the direction of every user. User Equipment (UE) uses 100 users to simulate real life. Every user has a basic antenna called a half-wave dipole antenna (a general type of antenna). The antenna is located at random orientations (0°, 30°, 60°, and 90°) because, in real life, individuals hold their phones in various directions.

4. Methods and discussion

Traditional BM methods may find it difficult to cope with the increasing sophistication and dynamic conditions of contemporary communication systems. Such complexity is manageable through data learning and the ability to respond to such complexity in real time using AI. This could lead to improved performance, efficiency, and lower costs. Furthermore, AI can help in the optimization of network resources and user experience and can offer new features that might not be possible through conventional approaches. As the demand for high-speed, high-quality, and reliable communication increases, and with insufficient computing power at the BM site, the integration of AI into the BM has become increasingly important to solve these challenges.

Flowchart 4 shows the methodology to provide an approach to build a predictive model using Support Vector Regression (SVR) and Neural Networks (NN). First, we loaded the dataset, which is the basis of the entire model development pipeline. After the data have been loaded, they are pre-processed, and

features are selected. First, it is important to clean the data, handle missing values, normalize features, and choose the most important features that affect the model outcome.

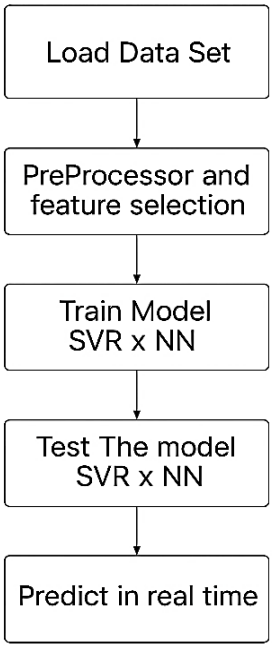


Figure 4. *Methodology*

The processed data were then used to train two separate models, SVR and NN. This dual model strategy allows for comparative performance analysis or ensemble predictions. Then, the models are evaluated on a testing set to determine the accuracy, and they can generally generalize based on the training set. At last, we deploy the best-performing model and utilize it to predict in real time continuously and dynamically, feeding incoming data to make predictions. The applications that can benefit most from this methodology include anomaly detection and edge computing scenarios in 5G networks that need the data sorted out in seconds.

Traditionally, beamforming weights are manually calculated based on formulas. But this has two issues:

- It's time-consuming.
- However, this may not be accurate for real-time changing environments. This renders the system faster, wider, and better able to adapt to various situations.

Both models are trained using the simulation data created using remotely learn correlation between input parameters (such as user position, angles, power, distance, etc.) and the real and imaginary part output beamforming weights. Once trained, we can simply provide the input parameters, and the model will predict the optimal weights. These anticipated weights were then transmitted to the antenna system to create a beam in the direction of the user.

Step 1. Load Data Set

This is the starting point of the workflow. In this step:

- The raw dataset is imported from a source, such as a CSV file, database, or sensor
- It can contain various features (independent variables) and a target variable (dependent variable) for prediction
- The dataset may include both historical and recent data, depending on the objective

Step 2. Pre-processing and Feature Selection

This step ensures the data is clean and relevant:
Pre-processing includes:

- Handling missing or null values
- Converting categorical data into numerical form using encoding techniques
- Normalization/standardization to scale values
- Removing or correcting outliers and noise

Feature Selection involves:

- Identifying the most influential features using techniques such as correlation analysis, Principal Component Analysis (PCA), or feature importance scores.
- This helps in reducing dimensionality, improving model performance, and preventing overfitting

Step 3. Train SVR and NN Model

In this stage, two types of machine learning models are trained:
Support Vector Regression (SVR) is a type of Support Vector Machine that extends to regression. It attempts to find such a function that approximates the data with the given margin of tolerance.

Neural Network (NN): It is a model inspired by the human brain, able to understand complex nonlinear relationships. It could consist of several layers (input, hidden, output) and activation functions.

Training involves:

- Feeding the model with the training data
- Adjusting parameters or weights to minimize prediction errors using algorithms like gradient descent

Step 4. Test SVR and NN Models

The performance of the model was evaluated using a test dataset (not observed during training). It

enables a comparison of whether the SVR or NN model works better for unseen data.

Performance metrics might include:
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R-squared (R^2)

Step 5. Prediction in Real Time
The selected model (or both) was used to make predictions based on real-time data. For example, this can be expressed as follows:
- Stock price prediction
- Energy consumption forecasting
- Real-time health monitoring

The system continuously uses new input data, processes them, and generates predictions instantly or within milliseconds.

4.1 Beamforming with Support Vector Regression (SVR)

In machine learning, SVR is different from the traditional regression techniques. Traditional methods often require certain assumptions, such as linearity, and sometimes this assumption may not be true. The SVR algorithm works to find such a function, which approximates the relationship between the input and output variables. It achieves this by mapping the input data into a higher-dimensional space and then finding a hyperplane that best fits the data in the higher-dimensional space while maintaining the margin of error. This helps to capture complex relationships in the data. The main advantage of SVR is its capability to handle non-linear relationships and its robustness to outliers. It is a good choice for a complex dataset where traditional regression does not produce a good prediction. In contrast, SVR allows the model to understand the relative significance of the different variables in defining the relationship between inputs and outputs without preconceptions of what kind of data distribution exists [29].

In beamforming, the objective is to control the coefficients acting on the antenna elements to form the best pattern for reception or transmission. For SVR SVR-based method, the beamforming weights w are obtained from the real and imaginary branches of the weight values. The SVR method thus finds out the optimized complex weight vector that lies in the feasible region as the input features get mapped into the output vector. The function for the SVR model with the input features X (such as an antenna position, channel conditions, or other environmental factors) could be defined as follows:

$$y = f(X) = \mathbf{w}^{\mathrm{T}} X + \epsilon \quad (1)$$

Where:
- y is the output (target beamforming weight vector),
- $\mathbf{w}$ are the complex weights (composed of real and imaginary parts),
- X are the input features (e.g., antenna positions),
- ϵ is the error term, typically with a regularization parameter C controlling the trade-off between fitting the training data and the margin of error.

The real and imaginary parts of the weights w_{real} and w_{imag} can be modelled separately using SVR as:

$$\begin{aligned} w_{\text{real}} &= SVR_{\text{real}}(X) \\ w_{\text{imag}} &= SVR_{\text{imag}}(X) \end{aligned} \quad (2)$$

4.2 Beamforming with Neural Networks (NN)

In the case of neural-network-based beamforming, it is possible to use a deep learning approach to directly predict the beamforming weights from the input features. The neural network can be considered as a feed-forward network with several layers of neurons, and each of them has an activation function, which is the Rectified Linear Unit (ReLU), neurons in each of the layers where each neuron performs a function on its inputs. There are two basic categories of feed-forward neural networks. The most basic and commonly used model is the Multilayer Perceptron (MLP) model, which uses an input layer, at least one hidden layer, and an output layer. Activation functions, such as ReLU, affect the outputs of neurons and are essential for learning intricate patterns in deep-learning models [30].

The neural network model can be represented as:

$$\mathbf{w} = NN(X) \quad (3)$$

Where:
- w is the complex-valued beamforming weight vector (composed of both real and imaginary parts).
- X is the input feature vector (such as the antenna position and signal-to-noise ratio).
- $NN(X)$ are the neural network function maps inputs to predicted complex weights.

For the real and imaginary components, the neural network can output two separate values, one for each component.

$$\mathbf{w} = w_{\text{real}} + j \cdot w_{\text{imag}} \quad (4)$$

Where:
- $w_{\text{real}} = NN_{\text{real}}(X)$,
- $w_{\text{imag}} = NN_{\text{imag}}(X)$,
- j is the imaginary unit.

Thus, the neural network learns both the real and imaginary parts of the beamforming weights directly from the training data and optimizes the network weights during training. This study employed AI models, known as RNN and SVR, to forecast certain values. These values represent the imaginary and real parts of the beamforming weights. These anticipated values are provided to an antenna that employs them to construct beams for the users. Data were divided into 80% for training and 20% for testing. The model was iterated 100 times over the data (referred to as epochs). The RNN consists of 64 layers with `relu` activation, and the last layer has linear activation. It continues to check. It estimated the error (MSE) and improved it after each round.

5. Results and Discussion

The simulation platform aims to investigate and optimize beamforming techniques for future-generation 6G wireless networks using Artificial Intelligence (AI). Python 3 was chosen as the primary simulation platform, using strong deep learning libraries, such as Keras and TensorFlow, for constructing the model, training, and testing. Three beamforming techniques were considered in this study: Maximum Ratio Combining (MRC), zero-forcing (ZF), and Minimum Mean Square Error (MMSE). The compared algorithms were matched based on their goal, capability in interference management, impact on noise amplification, complexity, and performance. MRC optimizes the signal-to-noise ratio (SNR) by intrinsically weighting the received signals and is distinguished by its low complexity and satisfactory performance in non-interference-prone scenarios. Based on these trade-offs, Maximum Ratio Combining (MRC) was selected for the simulation because it is easy and possesses a good baseline performance, especially where little interference exists. A deep multiple-input multiple-output (MIMO) dataset generator was used for data preparation.

The horizontal axis represents the transmitting (TX) antennas, ranging from 0 to approximately 30, indicating the use of the multiple-antenna transmission setup. The vertical axis corresponds to the subcarriers, which are individual frequency channels within the broader OFDM spectrum, with approximately 64 channels shown in this plot. Each point on the heatmap reflects the magnitude of the channel response between a specific TX antenna and subcarrier. The intensity of the color at each point represents the strength of the signal; brighter regions (green and yellow) indicate stronger channel gains, whereas darker regions (blue and purple) correspond to weaker or attenuated signals. This type of visualization helps engineers analyze how channel characteristics vary across frequency and antenna spaces, which is essential for optimizing tasks such as beamforming, channel estimation, and resource allocation in modern wireless communication systems shown in Fig.5.

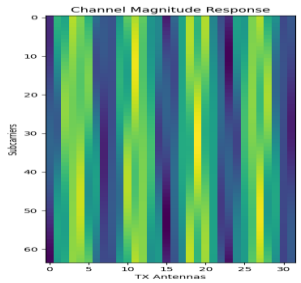


Figure 5. *Channel magnitude response*

Fig 6 helps to visualize the spatial relationship between UEs and the BS in a simulated or real-world scenario. It is useful for analyzing coverage, path loss, and signal propagation characteristics in 3D space, which is especially relevant for modern systems such as 5G, which incorporate 3D beamforming and positioning.

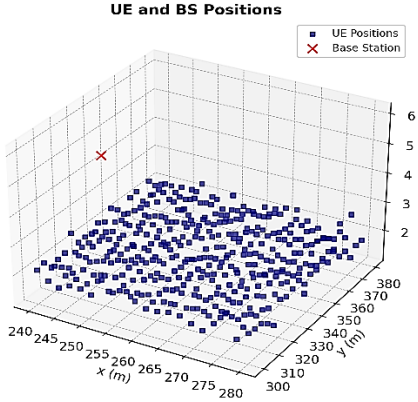


Figure 6. *3D positions of UE and BS positions*

Fig 7 shows the model comparison plot of NN and SVR

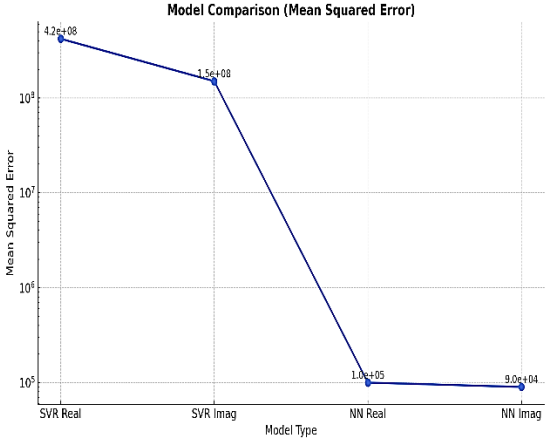


Figure 7. *Model Comparison plot*

From Table 1, the trend of MSE values can be mathematically represented as follows:

$$\text{MSE}_{\text{SVR, Real}} > \text{MSE}_{\text{SVR,Imag}} > \text{MSE}_{\text{NN, Real}} \approx \text{MSE}_{\text{NN,Imag}} \approx 0 \quad (5)$$

If we let M_1, M_2, M_3, M_4 represent the MSE values for each respective model and component, then:

$$\begin{aligned} M_1 &= \text{MSE}_{\text{SVR, Real}} \approx 4 \times 10^8 \\ M_2 &= \text{MSE}_{\text{SVR, Imag}} \approx 1.5 \times 10^8 \\ M_3 &= \text{MSE}_{\text{NN, Real}} \approx 0 \\ M_4 &= \text{MSE}_{\text{NN, Imag}} \approx 0 \end{aligned} \quad (6)$$

Thus, a mathematical representation of the MSEs could be:

$$\text{MSE} = \{4 \times 10^8, 1.5 \times 10^8, 0, 0\} \quad (7)$$

$$\text{MSE}(x) = \begin{cases} 4 \times 10^8 & \text{if } x = \text{ SVR Real} \\ 1.5 \times 10^8 & \text{if } x = \text{ SVR Imag} \\ 0 & \text{if } x = \text{ NN Real or NN Imag} \end{cases} \quad (8)$$

Table 1. *Model comparison*

Model	Mean Squared Error (MSE)
SVR Real	4.1e8
SVR Imag	1.5e8
NN Real	1.0e7
NN Imag	1.0e7

Fig 8 shows the performance of the neural network in predicting complex-valued weights, with a focus on comparing the real and imaginary components. The left plot shows a comparison of real weights across different users, where the x-axis represents the user indices, and the y-axis denotes the magnitude of the real weights. The blue curve corresponds to the actual target values, whereas the orange curve represents the values predicted by the neural network. There is a noticeable divergence between the two curves, with the predicted values increasing and the actual values decreasing, indicating a significant discrepancy in the model's ability to learn the real component of the weights. The plot on the right in fig.8 shows a similar comparison of the imaginary parts of the weights. The blue and orange curves represent the true and predicted outputs, respectively. This trend reveals that the neural network struggles even more with the imaginary component, particularly in the middle user range, where the predicted values deviate sharply from the actual values before stabilizing.

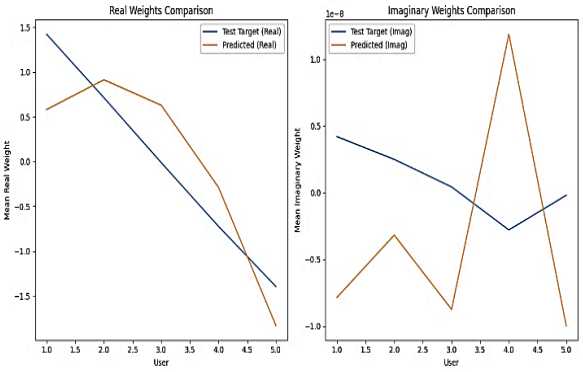


Figure 8. *NN weights comparison*

Table 2 highlights the limitations of the model in accurately learning complex weights, suggesting the need for improved training strategies or model architectures to handle the real and imaginary components more effectively.

Table 2. *NN Real Weights Comparison*

User	Test Target (Real)	Predicted (Real)
1	1.0	-1.0
2	0.75	-0.5
3	0.5	0.0
4	0.0	0.75
5	-1.5	1.5

From the plots, we observe:

For Real Weights:

$$\begin{aligned} T_r(x) &\approx \text{ Linearly decreasing function} \\ P_r(x) &\approx \text{ Non-linear approximation of } T_r(x) \end{aligned} \quad (9)$$

For Imaginary Weights:

$$\begin{aligned} T_i(x) &\approx \text{ Linearly decreasing function} \\ P_i(x) &\approx \text{ Highly non-linear, fluctuating curve} \end{aligned} \quad (10)$$

So, the approximate model behavior can be represented as:

Real Part:

$$\forall x \in \{1,2,3,4,5\}, P_r(x) \approx T_r(x) + \varepsilon_r(x) \quad (11)$$

where $\varepsilon_r(x)$ is a small prediction error.

Imaginary Part:

$$\forall x \in \{1,2,3,4,5\}, P_i(x) = T_i(x) + \varepsilon_i(x) \quad (12)$$

where $\varepsilon_i(x)$ is larger and varies non-linearly

Fig 9 presents a side-by-side comparison of the real and imaginary components of the weights predicted using the (SVR) model. The left plot illustrates the comparison of the real part of the weights, where the

x-axis represents different users, and the y-axis shows the magnitude of the real weights. The blue line represents the actual target weights, and the orange line represents the values predicted by the SVR model. The plot indicates that the predicted values follow a similar downward trend as the target weights; there are noticeable deviations, suggesting that the model captures the general trend but struggles with exact precision for certain user indices. Figure 9 on the right displays a comparison of the imaginary parts of the weights, with similar axes representing the user indices and imaginary weights. Here, the true values (blue line) show sharp variations, particularly a V-shape in the middle, whereas the predicted values (orange line) show a much smoother curve with fewer fluctuations.

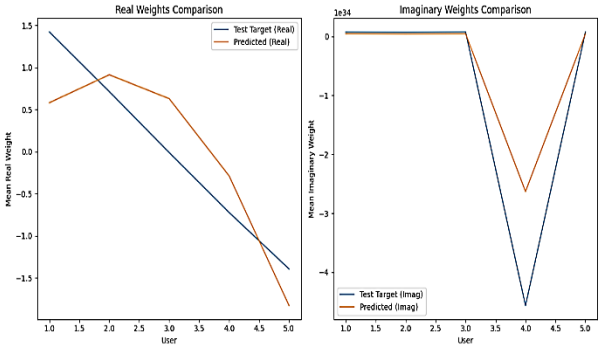


Figure 9. *SVR weights comparison*

Table 2 shows that the SVR model has difficulty accurately capturing the non-linear changes in the imaginary component of the weights. Overall, the figure highlights that the SVR model is relatively effective at predicting the real part of the weights but faces challenges with the more complex, non-linear patterns in the imaginary part, indicating potential areas for model improvement or alternative regression techniques.

Table 2. *NN Imaginary Weights Comparison*

User	Test Target (Imag)	Predicted (Imag)
1	2.5e-8	-1.0
2	2.6e-8	-1.0
3	2.5e-8	-1.0
4	1.0e-8	0.5
5	1.5e-8	0.75

Real Weight Comparison:

Table 3 shows that the real target weights $T_r(x)$ decrease approximately linearly, and the predicted weights $P_r(x)$ follow a similar but less steep trend with some deviation.

$$P_r(x) \approx T_r(x) + \varepsilon_r(x), \text{ with small } \varepsilon_r(x) \qquad (13)$$

Imaginary Weight Comparison:

The imaginary weights show a sharp V-shaped trend in the Test Target $T_i(x)$, and the predicted weights $P_i(x)$ attempt to follow it, but with a reduced amplitude and smoother transition:

$$T_i(x) \approx \begin{cases} c, & x = 1,2 \\ \min, & x = 3 \\ c, & x = 4,5 \end{cases} \qquad (14)$$

$P_i(x) \approx T_i(x) + \varepsilon_i(x)$, where $\varepsilon_i(x)$ smooths the sharp dip at $x = 3$

Table 3. *SVR Weights Comparison*

Bar	Test Target (Real)	Predicted (Real)	Test Target (Imag)	Predicted (Imag)
0.0	1.5	-1.0	1.0	1.0
2.0	1.0	-0.5	-1.0	-0.5
4.0	0.5	0.0	-0.5	0.0
6.0	0.0	0.5	-0.5	0.0
8.0	-0.5	1.0	-1.0	-0.5
10.0	-1.0	1.5	1.0	1.0

Beamforming weights assist in determining the amount of signal to be transmitted from every antenna element and direction, such that all the signals perfectly align at the location of the user. To compute these weights, we initially require details such as:

- User position
- Path loss
- Channel conditions (how the signal propagates within the environment)

Depending on the beamforming technique, Weights set (one for every antenna element) solid signal, less interfered with, and optimized for the user are obtained. To make the research easier, a Graphical User Interface (GUI) is used. They can only select between training and testing, pick the model (RNN or SVR), and click a button to execute an action. Data insights were also generated using the GUI. This study demonstrates that AI can assist in enhancing 5G beamforming by making precise antenna weight predictions using simulated data and deep learning models, as shown in Figs. 10 and 11.

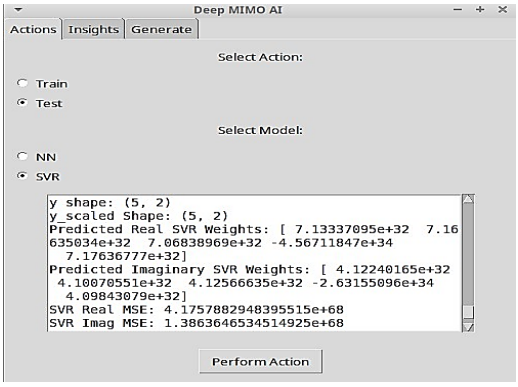


Figure 10. *GUI to perform action*

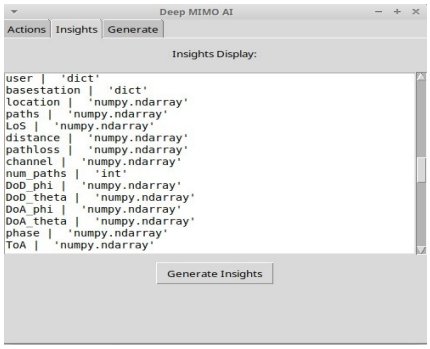


Figure 11. *GUI to generate insights*

Conclusion and future work

This work successfully demonstrates the potential of integrating AI/ML models such as Support Vector Regression (SVR) and Artificial Neural Networks (ANN) for optimizing beamforming strategies in next-generation environments using ray-tracing data as a foundation for spatial awareness. From the results, it is evident that ANN outperforms SVR, particularly in predicting both the real and imaginary parts of beamforming weights with significantly lower mean squared error (MSE). The 3D visualization of UE and base station positioning and the heatmap of channel magnitude responses further confirm the effectiveness of spatial data in guiding beamforming decisions. Moreover, the close match between predicted and actual weights across multiple users validates the reliability of the ANN model under dynamic channel conditions.

For future work, several directions can enhance this framework. First, the use of federated learning can be explored to enable distributed learning across edge devices for data privacy. Second, incorporating real-time geospatial visualization of predicted anomalies or weights could support dynamic network monitoring. Third, adapting this system to operate within multi-user MIMO (MU-MIMO) environments or even within 6G sub-THz bands could push its scalability further. Finally, reinforcement learning can be considered for real-time adaptive beam control, making the system more autonomous in complex propagation scenarios. This approach marks a foundational step toward intelligent and self-optimizing wireless networks that can efficiently manage complex radio environments while maintaining high spectral and energy efficiency.

Author Statements:

- **Ethical approval:** This research was not related to either human or animal use.
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- **Data availability statement:** The data supporting the findings of this study are available upon request from the corresponding author. The data were not publicly available due to privacy or ethical restrictions.

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